

# THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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## ABSTRACT

We investigate the manner in which physical scale and population size jointly condition the emergence of complex behavior within open-ended ecological environments. Within our setting, agents remain wholly unsupervised, being furnished with neither explicit rewards nor prescribed learning objectives; rather, they evolve over successive generations through the interacting dynamics of reproduction, mutation, and selection. In the course of acting, agents concurrently reshape both their surrounding environment and the composition of the population itself, thereby sustaining a continuously dynamic ecology. Our objective is not the optimization of a single high-performance policy, but instead the systematic examination of how behaviors arise and evolve across large populations under the influence of natural competition and environmental pressure. Leveraging contemporary hardware in conjunction with a novel multi-agent simulator, we scale both the environment and the population to magnitudes substantially exceeding those attempted in prior work, attaining populations in excess of 60,000 agents, each governed by its own independently evolved neural network policy. We identify a diversity of emergent behaviors—among them long-range resource extraction, vision-based foraging, and predation—that arise under the conjoint pressures of competition and survival. Examining how sensing modalities and environmental scale modulate the emergence of these behaviors, we find that certain of them materialize only within sufficiently large environments and populations, and, moreover, that increasing scale confers greater stability and consistency upon the emergent behaviors so obtained. While evolutionary settings possess a long and rich research tradition, our scaling results on modern hardware open promising new avenues for the study of ecology as an instrument of machine learning, precisely in an era characterized by increasingly abundant computational resources and efficient machine frameworks. Experimental code will be released upon acceptance.

## 1 INTRODUCTION

Ever since Darwin, the evolutionary emergence of biodiversity and complex behavior has constituted an object of sustained inquiry across a broad spectrum of biological disciplines, encompassing evolutionary theory (Darwin 1859; Simpson 1944), ecology (MacArthur & Wilson 2001), and genetics (Dobzhansky 1937). The formal characterizations of these emergent dynamics span a considerable range, extending from theories of speciation and adaptive radiation (Coyne & Orr 2004; Schluter 2000) through to the mathematical analyses of population genetics (Fisher 1930; Wright 1931).

Within these fields, decades of mathematical modeling, laboratory experimentation, and field observation have profoundly advanced our comprehension of the mechanisms governing evolution in nature. Regrettably, the investigation of emergent behavior in the natural world remains beset by numerous obstacles. As ecosystems grow in magnitude and complexity, they become correspondingly more resistant to controlled manipulation and precise measurement. Even where such manipulation is feasible, the conduct of controlled experiments within large-scale natural settings incurs the attendant risk of damaging or displacing wild populations.

With a view to elucidating more fully the emergence of complex behavior under competitive and environmental pressures, we undertake a study of the open-ended evolution of neural network poli-

cies within large-scale ecological simulations. In this setting, agents are endowed with neither a specified objective nor a reward signal; instead, they gather resources so as to survive and reproduce in accordance with the dynamics of the environment, and modifications to a policy occur exclusively through mutation as parents transmit their policies to their offspring. Given the rapid expansion of the computational resources available to researchers, our aim is to interrogate open-ended evolutionary dynamics within populations of these simple digital organisms at scales hitherto unattainable, thereby furnishing new instruments for investigating how environmental scale and population size bear upon behavioral emergence.

To this end, we employ a new JAX-based simulation environment that affords the rapid evaluation of expansive grid worlds. Our larger experiments accommodate upward of 60,000 individual agents distributed across a physical area exceeding 1,000,000 grid cells. Agents inhabiting this environment must navigate in search of food and other resources if they are to survive, and they may further reproduce by amassing sufficient resources to instantiate a mutated copy of themselves. A 3D visualization of this environment is provided in Figure I

Our aim is not merely to deepen our understanding of ecological phenomena through simulation, but additionally to investigate ecology itself as a mechanism for the production of machine intelligence. Recent years have made it abundantly clear that, within artificial intelligence, scale does not simply improve models—it fundamentally transforms their very capabilities (Wei et al., 2022a). Yet notwithstanding this insight, the majority of attempts at embodied learning have remained confined to comparatively small environments sustaining low populations. Our work interrogates what capabilities emerge when the object of scaling is not a single model but an entire ecosystem. Just as reasoning emerges within individual models only once a certain size threshold has been surpassed (Wei et al., 2022b), so too do we seek to understand which strategies evolve within populations only when environmental size and complexity attain sufficient richness.

Through an extensive suite of experiments, we identify multiple instances in which sensor configurations, in concert with our large environmental scales, exert a pronounced influence upon the behaviors emergent from simulation, an influence that may in turn engender long-term ecological consequences. For instance, we demonstrate that agents equipped with simple compass sensors can adapt so as to undertake long-range resource-gathering expeditions, yet that this behavior emerges with diminished reliability at small scales. We further demonstrate that agents furnished with vision sensors evolve superior foraging and attacking behavior relative to those lacking them; here again, these effects manifest more reliably at large scales. Taken collectively, these experiments establish that the scale and complexity afforded by modern hardware accelerators constitute promising new means for the study of how evolution within ecological environments can give rise to rich novel behaviors.

## 2 RELATED WORK

The emergence of life has served as a wellspring of inspiration for computational research since the field’s very inception. Conway’s Game of Life (Games, 1970) simulates populations by means of simple local rules. Early computer programs such as Tierra (Ray, 1992) and Avida (Brown, 1992) conceive of life as self-replicating programs that evolve through successive mutation. In a similar vein to Conway, Reynolds (1987) availed themselves of local rules of motion to elicit emergent flocking and schooling behaviors.

Researchers have marshalled these principles to evolve both the locomotion morphology (Sims, 1994a,b; Ha, 2018; Gupta et al., 2021; Auerbach et al., 2014; Auerbach & Bongard, 2012; Jeon et al., 2025; Auerbach & Bongard, 2014; Szerlip & Stanley, 2013) and the sensing apparatus (Pratt et al., 2022; Mugan & MacIver, 2020; Tiwary et al., 2025) of virtual agents, whereas neuroevolution methods have undertaken to evolve neural network architectures in a direct manner (Stanley & Miikkulainen, 2002; 2004; Stanley et al., 2009). In contradistinction to these approaches, we do not endeavor to produce agents capable of adapting more effectively to their environment by recourse to some reward or objective signal; rather, we seek to apprehend which behaviors emerge of their own accord as a consequence of ecological pressure, and to discern the impact that environmental scale and complexity exert upon these behaviors.

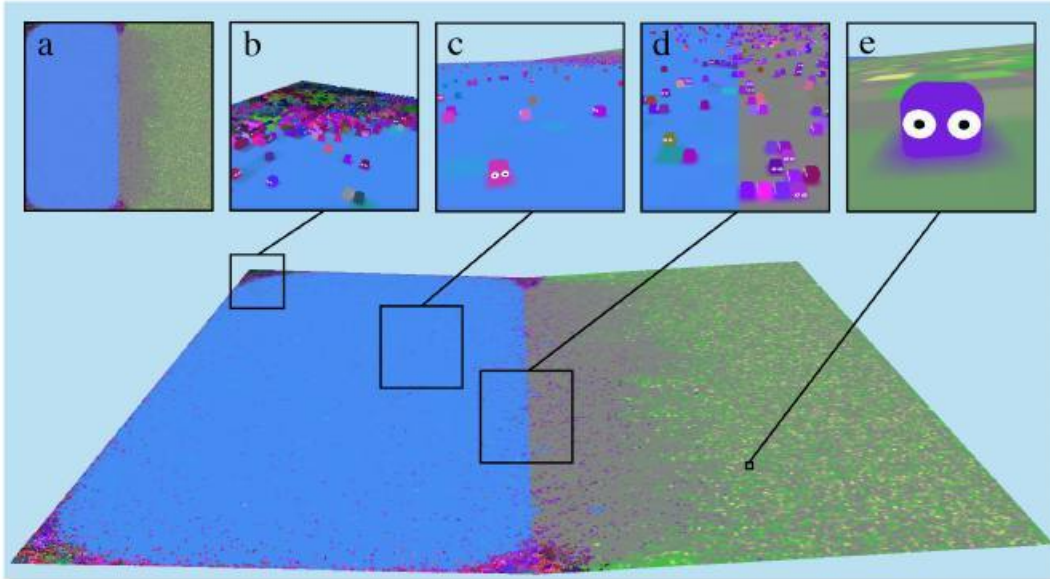


Figure 1: A 3D visualization of a  $512 \times 512$  map within our environment. Insert (a) presents a top-down view of the map in its entirety; (b) depicts a group of agents clustered in the corner; (c) displays agents situated in the water; (d) illustrates agents travelling back and forth between land and water; and (e) offers a close-up of a single agent in the act of gathering resources.

Evolution has likewise furnished inspiration for the internal dynamics of optimization algorithms. Genetic algorithms (Holland, 1992; Jong, 1975) and evolutionary strategies (Wierstra et al., 2014; Hansen & Ostermeier, 2001; Rechenberg, 1973; Schwefel, 1995; Conti et al., 2018) transpose the principles of evolution and natural selection onto the domain of optimization. Cognate works, in turn, deploy evolutionary methods so as to promote novelty among the resulting solutions (Lehman & Stanley, 2011; Cully et al., 2015). Still others have brought these approaches to bear upon the optimization of neural networks for tasks such as Atari games and Mujoco locomotion tasks (Salimans et al., 2017; Such et al., 2017; Mania et al., 2018). Although our work does employ evolutionary mechanisms in the updating of agents, our emphasis rests upon the study of emergent behavior rather than upon the development of a novel optimization algorithm or the deployment of an existing one toward the accomplishment of a fixed objective.

There likewise exists a rich body of work that draws upon biological inspiration in order to develop organisms situated within an environment. Early exemplars include Bedau (1992), Yeager et al. (1994), Channon & Damper (1998), and Tisue et al. (2004). Certain recent works avail themselves of cellular automata to grow organisms possessed of emergent morphologies and behaviors (Mordvintsev et al., 2020; Robinson, 2021; Heinemann, 2021; Randazzo & Mordvintsev, 2023; Caussan, 2025), while Mordatch & Abbeel (2018) and Park et al. (2023) enlist simulated environments to investigate the emergence of language and communication among agents. The emergence of behavior attributable to autocurricula (Leibo et al., 2019)—the notion that populations of agents furnish for one another an increasingly formidable learning landscape—has similarly been explored, though within environments hosting populations smaller than ours (Leibo et al., 2018; Team et al., 2021). In addition, Wang et al. (2019) and Aki et al. (2024) adopt the converse strategy, evolving environments specifically optimized for the ideal learning of agents. Although these works evolve the environment in place of the agent, they nonetheless inform our own, insofar as they lay bare the manner in which environmental complexity can elicit sophisticated behavior.

Most closely allied to our work are several efforts that simulate populations of agents interacting within complex environments. Baker et al. (2019) examine a Hide-and-Seek task in which a small contingent of hider agents learns to exploit environmental tools so as to evade seeker agents, both parties being trained via reinforcement learning. Other works comparably train populations toward specific ends: the playing of video games (Jaderberg et al., 2019), the manipulation of objects (Petrov et al., 2021), or the capture of agents (Zheng et al., 2017; Yamada et al., 2020). Yet however

much many of these works engage with emergence, they do so under the aegis of fixed tasks and objectives. The Neural MMO framework (Suarez et al. 2019, 2023) adopts a more open-ended posture toward reinforcement learning, situating agents within a massively multi-agent game setting and optimizing solely for survival in lieu of fixed objectives. The population, however, remains capped at a fixed number of agents (with experiments examining up to 128 concurrent agents) and devoid of generational inheritance, thereby constraining any examination of population dynamics. Hamon et al. (2023) aligns more closely with our work by virtue of its variable population sizes, genetically inspired evolution, and generational timelines, though its environment is markedly smaller, less feature-rich, and does not directly interrogate the significance of scale. In a similar spirit, Lu et al. (2024) likewise simulates agents that evolve to gather resources. This work, too, however, operates at a considerably smaller scale (a maximum of 256 agents) and places its emphasis upon tool use mediated by programmable robots. Although each of these antecedent works advances toward the examination of emergent ecological dynamics, a comprehensive understanding demands populations capable of growing and contracting of their own accord, of evolving in the absence of predefined objectives, and of interacting within environments sufficiently rich to support diverse strategies at scales at which collective patterns may emerge. Our work integrates these elements in concert, thereby enabling the examination of ecological phenomena that lay beyond the reach of prior work.

Finally, we draw further inspiration from recent works that foreground environmental speed and scale, harnessing optimized libraries so as to augment the computational throughput and the number of agents able to interact within game-like environments (Koyamada et al. 2023; Matthews et al. 2024; Lechner et al. 2023). We build upon this progress in computational efficiency, availing ourselves of analogous JAX-based acceleration techniques in order to achieve both large scales and rapid simulation within ecological environments.

### 3 METHOD

#### 3.1 ECOLOGICAL GAMES

We model the interactions obtaining between populations of agents and their environment as a species of multiplayer game that we designate an **Ecological Game (EG)**. In a manner analogous to such mathematical models as Markov decision processes (MDPs) in reinforcement learning and stochastic games (SGs) in multi-agent reinforcement learning (see Sutton et al. (1998) and Albrecht et al. (2024) for reference), an EG possesses underlying Markovian transition dynamics, yet it is expressly contrived to articulate populations of agents that change over time within an objective-free setting. In common with an SG, an EG comprises a set of agents  $I$ , a state space  $S$ , an initial state distribution  $\rho_0$ , an action space  $A$ , and a transition function  $\mathcal{T}$  that maps states  $s_t$  together with a set of actions for each player  $a_{t,i}$  to a distribution over subsequent states  $s_{t+1}$ . In departure from an SG, however, it dispenses with the reward function and introduces in its stead a population function  $\Gamma(s_t)$  that conveys which of the agents in  $I$  are presently alive as well as the identity of their parents, so that birth and death events may be inferred. Precisely as MDPs and SGs admit partially observable counterparts in the POMDP and POSG, so too may an EG be rendered partially observable, in which case we term it a POEG. The distinction between a POEG and an EG resides in the fact that a POEG withholds from each agent the capacity to observe the entire state, incorporating instead an observation function  $O(s_t, i)$  that maps a state and a player to an observation particular to that player. Our environment constitutes a POEG endowed with differing observation functions contingent upon the various sensor configurations elaborated in the subsequent subsection.

#### 3.2 ENVIRONMENT

The particular POEG employed in our experiments takes the form of a large grid world populated by various resources that agents must gather if they are to reproduce and survive. The environment sustains a factored state space  $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$ , wherein  $s_t^m$  denotes the 2D data associated with the map—the terrain and the distribution of resources, for instance—while  $s_t^a$  denotes the data associated with the agent, such as its position, its health, and the resources held internally.

**Terrain:** Each map is furnished with a rock layer that defines the height of every grid cell. Maps additionally incorporate water, which is initially deposited into those grid cells whose rock value falls beneath a preset sea-level. Over the course of an episode, water flows downhill by means of a discrete flow simulation predicated upon the height specified by this rock layer. In the experiments that follow, we make use of six global terrain configurations, displayed in Figure 2. **Ocean, Beach, Island, Lake, Isthmus, and Channel.** We conduct experiments at grid sizes of  $64 \times 64$ ,  $128 \times 128$ ,  $256 \times 256$ ,  $512 \times 512$ , and  $1024 \times 1024$ . Whenever the grid size is enlarged or diminished, we correspondingly scale the initial population size in proportion to the resulting change in the overall grid area. Once the simulation has commenced, the population thereafter fluctuates in accordance with the dynamics of the environment.

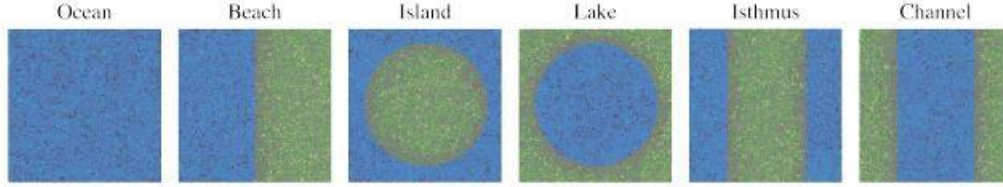


Figure 2: Overhead images of the terrains employed in our experiments at the  $256 \times 256$  grid size. Small dark spots denote individual agents, whereas green and yellow denote concentrations of energy and biomass, respectively. Each of these images was captured 5000 steps into an episode, at a juncture after the greater part of the biomass in the water has been consumed, yet before the agents have evolved so as to survive on land for more extended periods.

**Resources and Agent Health:** In order to survive and reproduce within these environments, agents must gather **water**, **energy**, and **biomass**. At the outset, biomass and energy are distributed at random throughout the environment, though both may thereafter be consumed and redistributed by individual agents.

Every action an agent undertakes expends a modest quantity of energy and water. The expended water is returned to the landscape, whereupon it flows downhill toward the nearest body of water, while the energy is consumed and thereby destroyed. Biomass, for its part, subserves reproduction; inasmuch as all agents require a fixed quantity of biomass merely to exist, they cannot reproduce until they have accumulated sufficient surplus biomass to bestow upon their offspring. As with water, the total biomass remains constant within the environment for the duration of the simulation; unlike water, however, biomass neither flows downhill nor moves of its own accord. Energy, by contrast, is not held constant but rather accrues slowly upon free-standing biomass. We contrived these resource dynamics so as to confront agents with a problem somewhat more exacting than the mere collection of a single food source. For ablations on resource rules, see Appendix F.2

Agents are further endowed with health points (HP) that they must maintain if they are to forestall death. Should an agent overspend a critical resource, its health deteriorates rapidly; conversely, it may expend energy and water so as to recover HP slowly over time. Agents may moreover attack one another, thereby diminishing the target’s HP. They additionally incur a small but monotonically increasing quantity of damage as a function of their age, with the consequence that agents ultimately succumb to old age if to no other cause. When an agent’s HP reaches zero and it dies, whatever resources it was carrying are returned to the environment within the grid cell upon which it stood. In this fashion, should an agent attack and kill another, it may thereupon consume the latter’s resources as a rudimentary form of predation. For ablations on HP hyperparameters, see Appendix F.8

**Sensors:** Agents apprehend their environment by means of four sensors. The **internal** sensor apprises the agent of its age, its health, and the resources it currently carries; the **external** sensor detects external resources residing within the same grid cell as the agent; a **compass**, inspired by the magnetoreceptors observed in birds and sundry other animals, registers the agent’s global orientation as a 4-way one-hot vector; and **vision** furnishes a  $7 \times 7$  top-down rendered patch surrounding the agent, comprising three color channels together with a fourth channel that encodes the local difference in elevation. For ablations on vision configurations, see Appendix F.6

**Actions:** Agents dispose of a finite set of discrete actions, partitioned into five categories. The first is **Rest**, which accomplishes nothing yet incurs the least energy. There are three **Move** actions: rotation clockwise, rotation counterclockwise, and forward advancement by one grid cell. Since multiple

agents cannot occupy a single cell, any move action is cancelled whenever an agent attempts to advance into a cell occupied by another, or whenever two agents attempt to move into the same cell. The agents additionally possess a single **Attack** action, which kills every agent situated within the  $3 \times 3$  square directly before them. For ablations on attack strength, see Appendix F.7. There are, furthermore, three separate **Eat** actions, one corresponding to each of the resource types. Should an agent occupy the same cell as a particular resource, the corresponding eat action transfers a fixed quantity of that resource from the environment into the agent’s stomach, which is subject to a fixed maximum capacity.

Finally, agents are afforded a single **Reproduce** action that begets a new offspring by way of single-parent reproduction. This action succeeds only in the event that the agent has amassed resources sufficient to bestow upon its new child. The newly created agent is instantiated directly behind its parent, its policy consisting of a mutated copy of the parent’s weights, and it immediately inherits a fixed quantity of the parent’s resources.

In the experiments that follow, we frequently disable certain sensors or the attack action so as to probe the ecological consequences of agents possessed of differing capabilities. We employ three agent sensing combinations: internal+external resources alone (**R**); internal+external+compass (**RC**); and internal+external+compass+vision (**RCV**). We adopt the notation **+A** to signify that attacking is permitted; thus, for example, (**RC+A**) designates an agent equipped with resource sensors and a compass that is moreover able to attack.

### 3.3 POLICIES

Our agent policies are implemented as small, memoryless MLPs. We elected for agents devoid of memory with a view to simplifying the computational model and curtailing the number of network parameters. The raw values issuing from each sensor are flattened, normalized to the range  $[-1, 1]$ , and passed through separate linear layers of output dimension 64. These values are subsequently summed and fed through a 2-layer MLP possessed of hidden dimension 64 and ReLU nonlinearities. Since the action space is discrete, the final output constitutes a linear layer whose output dimension equals the number of admissible actions. Actions are sampled from a softmax distribution whose temperature is permitted to evolve in concert with the network weights. Weights are represented by means of half-precision bfloat16 values, so as to accommodate larger population sizes and expedite run times. Contingent upon the number of sensors an agent employs, the network comprises, for each agent, roughly between 10k and 25k parameters. For ablations on network size and architecture, see Appendix F.5.

Across all runs, the weights for the initial population of agents are subject to Kaiming initialization (He et al. 2015), while bias vectors are initialized to zero. Policies for agents born in the midst of an episode are constituted by copying the parent’s weights and thereupon applying Gaussian noise of standard deviation  $3 \times 10^{-2}$  to all mutable weight values. This mutation scheme is congruent with that adopted in other neuroevolution research (Such et al. 2017). Although more elaborate approaches such as crossover are indeed possible, we opt for the present mechanism in order to simplify agent reproduction and thereby to study the effect of environmental scale within as unadorned a setting as is feasible. Empirically, we found a larger mutation rate to be necessitated by the use of low-precision floating-point numbers. We conduct an ablation on mutation rates in Appendix F.4.

Agents are additionally endowed with a 3-channel trait that governs their color upon being rendered onto the map. This color trait mutates in a manner analogous to the network weights and determines an agent’s appearance when observed by other agents equipped with the vision sensor. In principle, such a mechanism permits agents to distinguish among the several types of other agents.

### 3.4 COMPUTATION AND REPRODUCIBILITY

The entirety of our experiments is implemented in JAX (Bradbury et al. 2018) and executed upon Nvidia H100 and A40 GPUs within a state-of-the-art simulation environment. Unless otherwise noted, every experiment is run for 2M environment steps or until extinction supervenes, with per-seed wall-clock times ranging from under an hour for  $64 \times 64$  grids to approximately 10 hours for  $1024 \times 1024$  grids on a single H100 GPU. Appendix E sets forth the simulator architecture in greater

detail, reports additional scaling measurements, and canvasses considerations pertaining to reproducibility.

## 4 EXPERIMENTS

In the experiments that follow, we demonstrate the effects that differing sensing modalities, environmental scale, and population size exert upon the emergent behavior of agents.

### 4.1 LONG-DISTANCE RESOURCE GATHERING

In our first experiment, we examine the emergent capacity of agents to traverse considerable distances inland in the pursuit of resources. We demonstrate that, within environments possessed of clear directional structure, agents equipped with a compass (**RC**) consistently develop markedly more effective long-range resource-gathering behavior than does a baseline agent class (**R**) bereft of a compass. We further show that the emergence of this behavior grows more consistent and more effective as the environmental scale increases.

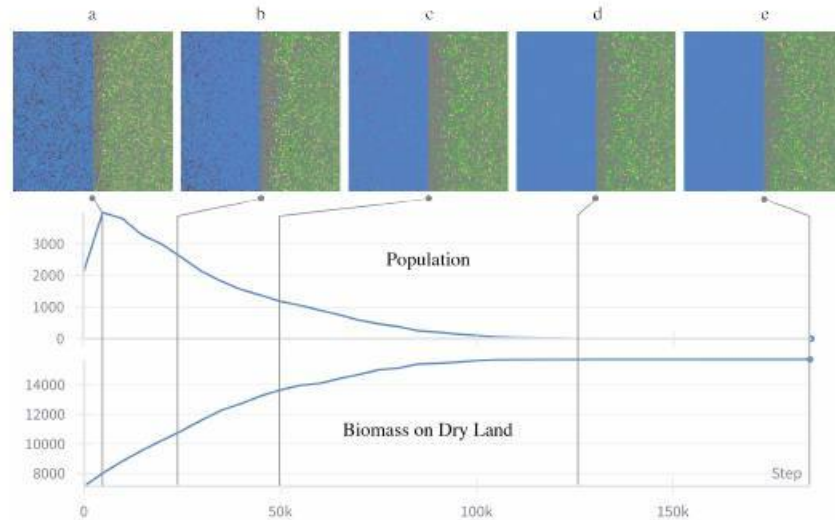


Figure 3: A representative trial involving (**R**) agents within a  $256 \times 256$  **Beach** environment. The upper chart traces population over time, whereas the lower chart traces the free biomass residing on dry land.

We begin by discussing a visual example of the (**R**) agent class, presented in Figure 3. By 5000 time steps (a), all of the initial agents situated on dry land have perished for want of water; in the interim, those residing in the water have proliferated rapidly on the strength of the rich initial resources. At this incipient stage, the population does not yet harbor sophisticated policies capable of efficiently seeking out resources. From this early demographic peak, agents in the water occasionally stray onto land (b); when they do, they are frequently unable to retrace their way back to the ocean and consequently perish, thereby depositing their biomass upon the land. By 50,000 time steps (c), this gradual process has very nearly doubled the quantity of uneaten biomass on land. This transfer of resources from water to land has, in turn, precipitated a contraction of the population, inasmuch as the biomass remaining in the water no longer suffices to sustain a large population. By 125,000 steps (d), almost the entirety of the biomass has been transferred to land, and the population verges upon extinction; the final agent expires at approximately 185,000 steps (e).

This example lays bare the ecological ramifications of the agents' inability to find their way back to water. Not only do the individuals themselves perish, but their deaths reconfigure the distribution of resources throughout the environment in ways that carry momentous consequences for the population in its entirety. At the same time, this ecological catastrophe furnishes an opportunity for any

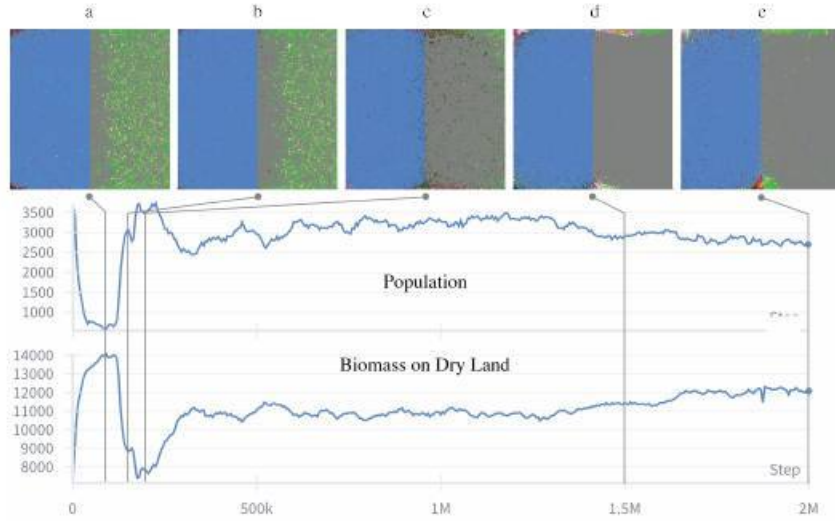


Figure 4: A representative trial involving **(RC)** agents within a  $256 \times 256$  **Beach** environment. The upper chart traces population over time, whereas the lower chart traces the free biomass residing on dry land.

agent capable of successfully discovering how to venture onto land, forage within the rich deposits of biomass bequeathed by its deceased forebears, and thereafter return safely to water.

In the ensuing example, depicted in Figure 4 we endow agents with an internal compass **(RC)**. Initially, a comparable pattern obtains (a), wherein the population contracts and the free biomass on land increases rapidly. In this instance, however, certain agents contrive to exploit their compass so as to drift into the relative safety of the corners. At approximately 150,000 steps (b), some agents commence to “mine” the biomass on land, availing themselves of their compass to navigate back and forth between land and water. By 200,000 steps (c), these miners have cleared the beach in its entirety. Thereafter the population undergoes only minor fluctuations, with agents coalescing into clumps about the edges (d), yet it remains largely stable throughout the full 2,000,000 time steps (e).

Table 1: Mining events within the **Beach** world in populations of agents endowed with a compass **(RC)**, contrasted with populations lacking one **(R)**, across five world sizes and initial populations. denotes the number of seeds that exhibited a mining event, whereas denotes the number of seeds in which the population went extinct before 2M steps. See Table 3 in Appendix A for detailed data pertaining to each seed.

|             | 1024 × 1024 | 512 × 512 | 256 × 256 | 128 × 128 | 64 × 64 |
|-------------|-------------|-----------|-----------|-----------|---------|
|             | 32768       | 8192      | 2048      | 512       | 128     |
|             |             |           |           |           |         |
| <b>(RC)</b> | 4/4         | 0/4       | 4/4       | 0/4       | 3/4     |
| <b>(R)</b>  | 1/4         | 0/4       | 0/4       | 4/4       | 0/4     |

We render these observations quantitative by conducting a comprehensive suite of **Beach** experiments across grid sizes of  $1024 \times 1024$ ,  $512 \times 512$ ,  $256 \times 256$ ,  $128 \times 128$ , and  $64 \times 64$ , with corresponding initial populations of 32768, 8192, 2048, 512, and 128 agents. Each experiment is run under four random seeds, and we record whether or not mining behavior emerges—which we operationalize as a decline of at least 10% in the free biomass on dry land relative to its historical maximum. We further record the onset time of the mining event, determined by the local maximum in biomass on land, together with the percentage decline in biomass. A summary of mining and extinction events is furnished in Table 1 with the full particulars for each seed relegated to Table 3 in Appendix A. Mining behavior emerges consistently among **(RC)** agents in contrast to **(R)**

agents. Moreover, mining emerges more consistently within larger environments, materializing in every seed from the  $256 \times 256$  scale upward; at smaller scales, by contrast, populations grow less stable and exhibit a heightened propensity toward earlier extinction.

To scrutinize these effects more closely, we run additional seeds with **(R)** and **(RC)** agents at the  $512 \times 512$  world size (32 seeds) and the  $128 \times 128$  world size (64 seeds), employing a Kaplan-Meier estimator to gauge the probability of extinction and of mining as a function of simulator time. We find that extinction is both rarer and slower to transpire at larger world sizes, whereas mining is more frequent and more prompt at smaller grid sizes. We additionally fit a Cox proportional hazards model in order to compare the timing of mining discovery and the timing of extinction across world sizes and compass usage, likewise finding that a larger world size substantially attenuates the rate of extinction and affords far more rapid mining discovery than obtains in smaller worlds; enabling the compass yields an effect of comparable magnitude. Full details are provided in Appendix [B](#).

Table 2: Mining and extinction events within the **Island**, **Lake**, **Isthmus**, and **Channel** terrains in populations of agents endowed with a compass (**RC**), contrasted with those lacking one (**R**), at the  $512 \times 512$  grid size.  denotes the number of seeds that exhibited a mining event, whereas  denotes the number of seeds in which the population went extinct before 2M steps. See Table [4](#) in Appendix [A](#) for detailed data pertaining to each seed.

|             | Island                                                                            |                                                                                   | Lake                                                                              |                                                                                   | Isthmus                                                                           |                                                                                   | Channel                                                                           |                                                                                     |
|-------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|             |  |  |  |  |  |  |  |  |
| <b>(RC)</b> | 4/4                                                                               | 0/4                                                                               | 2/4                                                                               | 2/4                                                                               | 4/4                                                                               | 0/4                                                                               | 3/4                                                                               | 1/4                                                                                 |
| <b>(R)</b>  | 3/4                                                                               | 1/4                                                                               | 2/4                                                                               | 4/4                                                                               | 1/4                                                                               | 3/4                                                                               | 0/4                                                                               | 4/4                                                                                 |

We conduct analogous experiments within the **Island**, **Lake**, **Isthmus**, and **Channel** terrains at the  $512 \times 512$  grid size with **(R)** and **(RC)** agents, with a view to investigating the bearing that terrain configuration has upon the emergence of mining behavior. Mining and extinction events for these experiments are reported in Table [2](#) while detailed information concerning each run is deferred to Table [4](#) in Appendix [A](#). As before, endowing agents with a compass conduces to the earlier and more consistent emergence of mining behavior, as well as to more stable populations attended by lower extinction rates. **(RC)** agents adapt more readily to the island and isthmus environments than to the lake and channel, within which they occasionally succumb to extinction. We conjecture that this disparity arises because walking due east or due west will invariably return an agent to water within the island and isthmus, whereas no comparably simple rule presents itself in the case of the channel and lake. The **(R)** agent, for its part, cannot reliably adapt to any of these environments, though it fares least poorly on the island. See Figure [12](#) in Appendix [D](#) for a visualization of mining within the island terrain.

## 4.2 VISUAL FORAGING AND PREDATION

We further conduct experiments within the **Ocean** environment in order to ascertain whether agents can adapt so as to exploit vision sensors in the effective location of free resources or the predation upon other agents. We contrast the performance of **(RCV+A)** agents—endowed with resource sensing, a compass, and a  $7 \times 7$  visual sensor—against that of **(RC+A)** agents possessed of resource and compass sensors alone.

Figure [5](#) plots population, biomass utilization, movement actions, and eat actions for agents within the  $512 \times 512$  and  $128 \times 128$  grid sizes. Plots for the remaining grid sizes are provided in Figure [9](#) in Appendix [C](#) while summary statistics for these plots are set forth in Table [5](#) in Appendix [A](#). Biomass utilization is computed as the percentage of the total biomass in the environment presently contained within the agents. We find that, in episodes wherein agents are furnished with vision sensors (red lines), they devote less time to movement and more to eating. These runs, moreover, sustain larger populations and attain greater biomass utilization. Such dynamics intimate that the agents have adapted so as to marshal their visual sensing capabilities toward foraging more efficiently for food. Upon comparison across environmental scales, we find that the variance between runs diminishes

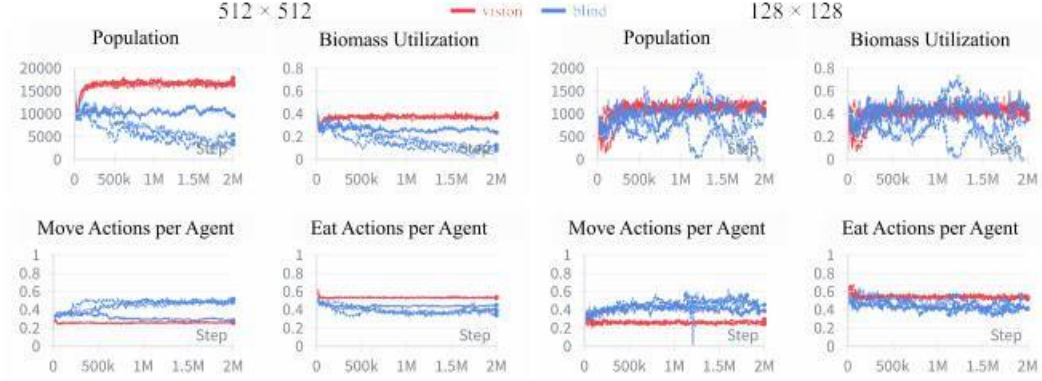


Figure 5: Population, biomass utilization, move actions per agent, and eat actions per agent within  $512 \times 512$  and  $128 \times 128$  **Ocean** worlds. Biomass utilization, move actions per agent, and eat actions per agent are averaged across agents at each time step. Each plot displays four seeds with vision agents (**RCV + A**) (red) alongside four seeds with blind agents (**RC + A**) (blue). When endowed with vision, agent populations prove larger and more stable and attain greater biomass utilization, a consequence of diminished movement together with more frequent and more effective eating. At larger scales these dynamics grow more stable, exhibiting less variance across runs, which intimates that within larger environments the vision sensor confers an ever-increasing advantage over blindness in the adaptation of foraging behavior. See Figure 9 in Appendix C for a comparison of grid sizes  $1024 \times 1024$  and  $256 \times 256$ .

substantially within larger environments. Unexpectedly, runs incorporating vision sensors exhibit markedly more stable dynamics, with attenuated variation in these metrics over time.

Figure 6 displays the averaged attacks per agent and homicides per attack for these runs. These statistics reveal that agents endowed with vision attack comparatively infrequently, yet sustain a successful kill rate of approximately 75%. Within large grids, agents lacking vision occasionally attain high kill rates as well, albeit under considerably more unstable dynamics. Regrettably, confounding factors render it difficult to adjudicate the underlying causes of this observation. By way of visual inspection (see Figure 13 in Appendix D), we have found that such high kill rates occasionally arise when clusters of agents accumulate at a single location, and that these clusters scarcely ever form in runs involving visual agents. This may in turn indicate that agents endowed with vision are able to adapt so as to avoid one another. Within smaller environments, the disparity between visual and blind agents grows less pronounced, though populations of visual agents remain far more stable.

#### 4.3 ABLATIONS

In Appendix F we undertake ablations upon specific hyperparameters—namely initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8)—so as to gauge the sensitivity of emergent behaviors to the simulator configuration. In addition, we conduct experiments in Appendix F.3 in order to verify that the dynamics observed within larger environments are not attributable merely to self-averaging.

### 5 CONCLUSION

Our study foregrounds the promise of open-ended ecological settings as a powerful paradigm for the investigation of the emergence of intelligence. Through experiments conducted in worlds of varying scale and complexity, we find that environmental scale and sensory richness alike contribute substantially to the development of sophisticated sensing and decision-making policies. Taken in aggregate, these results intimate that ecological and evolutionary pressures, when conjoined with sufficiently rich environments, may furnish a natural pathway toward the development of increasingly intelligent artificial agents. Our experiments demonstrate, furthermore, that rich behaviors

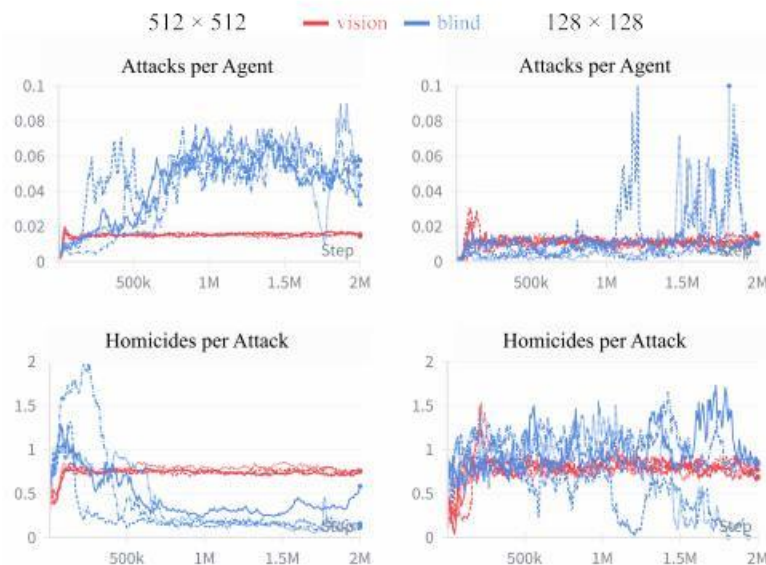


Figure 6: Attacks per agent and homicides per attack, averaged across agents, within 512×512 and 128×128 **Ocean** worlds. Each plot displays four seeds with vision agents (**RCV + A**) (red) alongside four seeds with blind agents (**RC + A**) (blue). When endowed with vision, agents attack less frequently yet with considerably greater precision, a pattern suggestive of the emergence of specialized predation behaviors. This dynamic grows more pronounced at larger scales. See Figure 10 in Appendix C for a comparison of grid sizes 1024×1024 and 256×256.

can be evolved within large populations under a reasonable hardware budget and without recourse to explicit objectives. We trust that these results will facilitate further exploration of ecology as a mechanism for the development and the understanding of artificial intelligence.

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## A ADDITIONAL DATA

Table 3: Mining events within the **Beach** world in populations of agents endowed with a compass (**RC**), contrasted with populations lacking one (**R**), across five scales.  denotes whether a run exhibited a mining event.  denotes the percentage decrease in free biomass on dry land over the course of the mining event, whereas  denotes the approximate timestep at which the free dry biomass attained the local maximum marking the onset of the mining event. For runs featuring multiple mining events, we report the first.  denotes the timestep at which the run went extinct, or n/a if it endured for the full 2M steps.

|                 | 1024x1024                                                                         |                                                                                   |                                                                                   |                                                                                   | 512x512                                                                           |                                                                                   |                                                                                   |                                                                                   | 256x256                                                                           |                                                                                   |                                                                                   |                                                                                   | 128x128                                                                           |                                                                                   |                                                                                   |                                                                                   | 64x64 |  |  |  |
|-----------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------|--|--|--|
|                 | 32768                                                                             |                                                                                   |                                                                                   |                                                                                   | 8192                                                                              |                                                                                   |                                                                                   |                                                                                   | 2048                                                                              |                                                                                   |                                                                                   |                                                                                   | 512                                                                               |                                                                                   |                                                                                   |                                                                                   | 128   |  |  |  |
| Compass On (RC) |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |       |  |  |  |
| Seed            |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |
| 0               | ✓                                                                                 | 35k                                                                               | 11.1                                                                              | n/a                                                                               | ✓                                                                                 | 105k                                                                              | 33.                                                                               | n/a                                                                               | ✓                                                                                 | 145k                                                                              | 49.3                                                                              | n/a                                                                               | ✓                                                                                 | 105k                                                                              | 72.5                                                                              | n/a                                                                               | x     |  |  |  |
| 1               | ✓                                                                                 | 90k                                                                               | 16.2                                                                              | n/a                                                                               | ✓                                                                                 | 50k                                                                               | 17.2                                                                              | n/a                                                                               | ✓                                                                                 | 50k                                                                               | 46.6                                                                              | n/a                                                                               | x                                                                                 |                                                                                   | 140                                                                               | ✓                                                                                 | 70k   |  |  |  |
| 2               | ✓                                                                                 | 40k                                                                               | 15.                                                                               | n/a                                                                               | ✓                                                                                 | 30k                                                                               | 14.3                                                                              | n/a                                                                               | ✓                                                                                 | 55k                                                                               | 34.7                                                                              | n/a                                                                               | ✓                                                                                 | 50k                                                                               | 50.7                                                                              | n/a                                                                               | ✓     |  |  |  |
| 3               | ✓                                                                                 | 60k                                                                               | 15.                                                                               | n/a                                                                               | ✓                                                                                 | 45k                                                                               | 15.6                                                                              | n/a                                                                               | ✓                                                                                 | 90k                                                                               | 47.4                                                                              | n/a                                                                               | ✓                                                                                 | 20k                                                                               | 52.                                                                               | n/a                                                                               | x     |  |  |  |
| Avg             | 4/4                                                                               | 34k                                                                               | 14.3                                                                              | 0/4                                                                               | 4/4                                                                               | 58k                                                                               | 20.                                                                               | 0/4                                                                               | 4/4                                                                               | 85k                                                                               | 44.5                                                                              | 0/4                                                                               | 3/4                                                                               | 58k                                                                               | 58.4                                                                              | 1/4                                                                               | 2/4   |  |  |  |
| Compass Off (R) |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |       |  |  |  |
| Seed            |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |
| 0               | x                                                                                 |                                                                                   |                                                                                   | n/a                                                                               | x                                                                                 |                                                                                   |                                                                                   | 945k                                                                              | x                                                                                 |                                                                                   |                                                                                   | 185k                                                                              | x                                                                                 |                                                                                   |                                                                                   | 65k                                                                               | x     |  |  |  |
| 1               | ✓                                                                                 | 275k                                                                              | 12.4                                                                              | n/a                                                                               | x                                                                                 |                                                                                   |                                                                                   | 1085k                                                                             | x                                                                                 |                                                                                   |                                                                                   | 110k                                                                              | x                                                                                 |                                                                                   |                                                                                   | 55k                                                                               | x     |  |  |  |
| 2               | x                                                                                 |                                                                                   |                                                                                   | n/a                                                                               | x                                                                                 |                                                                                   |                                                                                   | 1070k                                                                             | x                                                                                 |                                                                                   |                                                                                   | 165k                                                                              | x                                                                                 |                                                                                   |                                                                                   | 55k                                                                               | x     |  |  |  |
| 3               | x                                                                                 |                                                                                   |                                                                                   | n/a                                                                               | x                                                                                 |                                                                                   |                                                                                   | 280k                                                                              | x                                                                                 |                                                                                   |                                                                                   | 125k                                                                              | x                                                                                 |                                                                                   |                                                                                   | 90k                                                                               | x     |  |  |  |
| Avg             | 1/4                                                                               | 275k                                                                              | 12.4                                                                              | 0/4                                                                               | 0/4                                                                               |                                                                                   |                                                                                   | n/a                                                                               | 0/4                                                                               |                                                                                   |                                                                                   | 4/4                                                                               | 0/4                                                                               |                                                                                   |                                                                                   | 4/4                                                                               | 0/4   |  |  |  |

Table 4: Mining events within the **Island**, **Lake**, **Isthmus**, and **Channel** terrains among (RC) agents, contrasted with (R) agents. The grid size is  $512 \times 512$  with an initial population of 8192.

|                 | Island |      |      |      | Lake |     |      |       | Isthmus |     |      |      | Channel |     |      |      |
|-----------------|--------|------|------|------|------|-----|------|-------|---------|-----|------|------|---------|-----|------|------|
| Compass On (RC) |        |      |      |      |      |     |      |       |         |     |      |      |         |     |      |      |
| Seed            |        |      |      |      |      |     |      |       |         |     |      |      |         |     |      |      |
| 0               | ✓      | 40k  | 94.3 | n/a  | x    |     |      | 45k   | ✓       | 45k | 33.  | n/a  | ✓       | 50k | 27.4 | n/a  |
| 1               | ✓      | 30k  | 94.3 | n/a  | ✓    | 30k | 22.5 | n/a   | ✓       | 55k | 37.3 | n/a  | ✓       | 80k | 30.8 | n/a  |
| 2               | ✓      | 30k  | 94.8 | n/a  | x    |     |      | 45k   | ✓       | 40k | 28.1 | n/a  | ✓       | 50k | 41.  | n/a  |
| 3               | ✓      | 70k  | 95.1 | n/a  | ✓    | 35k | 25.3 | n/a   | ✓       | 30k | 41.5 | n/a  | x       |     |      | 80k  |
| Avg             | 4/4    | 43k  | 94.6 | 0/4  | 2/4  | 33k | 23.9 | 2/4   | 4/4     | 43k | 35.  | 0/4  | 3/4     | 60k | 33.1 | 1/4  |
| Compass Off (R) |        |      |      |      |      |     |      |       |         |     |      |      |         |     |      |      |
| Seed            |        |      |      |      |      |     |      |       |         |     |      |      |         |     |      |      |
| 0               | x      |      |      | 185k | x    |     |      | 125k  | x       |     |      | 145k | x       |     |      | 100k |
| 1               | ✓      | 75k  | 30.9 | n/a  | x    |     |      | 130k  | x       |     |      | 260k | x       |     |      | 105k |
| 2               | ✓      | 85k  | 47.  | n/a  | ✓    | 75k | 23.4 | 1400k | ✓       | 95k | 18.8 | n/a  | x       |     |      | 105k |
| 3               | ✓      | 115k | 39.5 | n/a  | ✓    | 70k | 24.3 | 1250k | x       |     |      | 140k | x       |     |      | 100k |
| Avg             | 3/4    | 92k  | 39.1 | 1/4  | 2/4  | 73k | 23.9 | 4/4   | 1/4     | 95k | 18.8 | 3/4  | 0/4     |     |      | 4/4  |

Table 5: Population and foraging data for Vision and Blind agents within the **Ocean** world at differing scales. denotes population size. denotes biomass utilization, namely the percentage of the total biomass in the system presently contained within the agents' stomachs. denotes move actions per agent, namely the fraction of time an agent took a move action, averaged across agents. In like manner, denotes the proportion of agents' actions that constituted an eat action. The reported values of , , , and are averages of the respective metric over all steps for the given seed.

|                | 1024x1024<br>32768                                                                  |                                                                                     |                                                                                     |                                                                                     | 512x512<br>8192                                                                     |                                                                                     |                                                                                     |                                                                                     | 256x256<br>2048                                                                     |                                                                                     |                                                                                      |                                                                                       | 128x128<br>512                                                                        |                                                                                       |                                                                                       |                                                                                       |
|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Vision (RCV+A) |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
| Seed           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0              | 58354                                                                               | .396                                                                                | .402                                                                                | .426                                                                                | 16413                                                                               | .369                                                                                | .257                                                                                | .532                                                                                | 4195                                                                                | .382                                                                                | .255                                                                                 | .536                                                                                  | 1126                                                                                  | .426                                                                                  | .261                                                                                  | .541                                                                                  |
| 1              | 65107                                                                               | .365                                                                                | .257                                                                                | .529                                                                                | 16200                                                                               | .364                                                                                | .256                                                                                | .533                                                                                | 4142                                                                                | .377                                                                                | .258                                                                                 | .536                                                                                  | 1103                                                                                  | .415                                                                                  | .262                                                                                  | .537                                                                                  |
| 2              | 64735                                                                               | .363                                                                                | .257                                                                                | .530                                                                                | 16560                                                                               | .372                                                                                | .257                                                                                | .533                                                                                | 4269                                                                                | .388                                                                                | .260                                                                                 | .536                                                                                  | 1041                                                                                  | .392                                                                                  | .254                                                                                  | .542                                                                                  |
| 3              | 64595                                                                               | .362                                                                                | .256                                                                                | .532                                                                                | 16206                                                                               | .365                                                                                | .255                                                                                | .531                                                                                | 4249                                                                                | .388                                                                                | .259                                                                                 | .533                                                                                  | 1100                                                                                  | .414                                                                                  | .257                                                                                  | .538                                                                                  |
| Avg            | 63198                                                                               | .371                                                                                | .293                                                                                | .504                                                                                | 16345                                                                               | .368                                                                                | .256                                                                                | .532                                                                                | 4213                                                                                | .384                                                                                | .258                                                                                 | .535                                                                                  | 1092                                                                                  | .412                                                                                  | .259                                                                                  | .539                                                                                  |
| Blind (RC+A)   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
| Seed           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0              | 22734                                                                               | .143                                                                                | .446                                                                                | .372                                                                                | 6549                                                                                | .173                                                                                | .442                                                                                | .400                                                                                | 2036                                                                                | .209                                                                                | .433                                                                                 | .461                                                                                  | 545                                                                                   | .219                                                                                  | .465                                                                                  | .427                                                                                  |
| 1              | 23670                                                                               | .151                                                                                | .426                                                                                | .388                                                                                | 6939                                                                                | .180                                                                                | .438                                                                                | .396                                                                                | 3107                                                                                | .321                                                                                | .390                                                                                 | .416                                                                                  | 1082                                                                                  | .444                                                                                  | .416                                                                                  | .460                                                                                  |
| 2              | 24745                                                                               | .159                                                                                | .358                                                                                | .426                                                                                | 5448                                                                                | .136                                                                                | .474                                                                                | .374                                                                                | 1587                                                                                | .163                                                                                | .440                                                                                 | .433                                                                                  | 979                                                                                   | .421                                                                                  | .404                                                                                  | .436                                                                                  |
| 3              | 20004                                                                               | .130                                                                                | .469                                                                                | .359                                                                                | 10354                                                                               | .265                                                                                | .311                                                                                | .439                                                                                | 3164                                                                                | .324                                                                                | .321                                                                                 | .450                                                                                  | 788                                                                                   | .317                                                                                  | .431                                                                                  | .477                                                                                  |
| Avg            | 22788                                                                               | .146                                                                                | .425                                                                                | .386                                                                                | 7323                                                                                | .189                                                                                | .416                                                                                | .402                                                                                | 2473                                                                                | .254                                                                                | .396                                                                                 | .440                                                                                  | 848                                                                                   | .351                                                                                  | .429                                                                                  | .450                                                                                  |

Table 6: Predation data for Vision and Blind agents within the **Ocean** world at differing scales.  denotes attacks per agent, namely the fraction of time an agent took the attack action, averaged across all agents.  denotes homicides per attack, namely the number of agents killed per attack action. The reported values of  and  are averages over all steps for the given seed. We draw attention to the outlier of 0.562 for  of seed 3 in the  $128 \times 128$  (**RC+A**) run: this seed went extinct prior to 2M steps, whereupon the dwindling population size occasioned a spike in the attacks-per-agent metric.

|                | 1024x1024                                                                           |                                                                                     | 512x512                                                                             |                                                                                     | 256x256                                                                             |                                                                                     | 128x128                                                                              |                                                                                       |
|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                | 32768                                                                               |                                                                                     | 8192                                                                                |                                                                                     | 2048                                                                                |                                                                                     | 512                                                                                  |                                                                                       |
| Vision (RCV+A) |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |
| Seed           |  |  |  |  |  |  |  |  |
| 0              | .011                                                                                | .776                                                                                | .015                                                                                | .731                                                                                | .013                                                                                | .784                                                                                | .010                                                                                 | .788                                                                                  |
| 1              | .016                                                                                | .733                                                                                | .015                                                                                | .741                                                                                | .014                                                                                | .780                                                                                | .011                                                                                 | .793                                                                                  |
| 2              | .016                                                                                | .707                                                                                | .015                                                                                | .737                                                                                | .014                                                                                | .750                                                                                | .012                                                                                 | .771                                                                                  |
| 3              | .016                                                                                | .712                                                                                | .015                                                                                | .780                                                                                | .013                                                                                | .759                                                                                | .011                                                                                 | .765                                                                                  |
| Avg            | .015                                                                                | .732                                                                                | .015                                                                                | .747                                                                                | .014                                                                                | .758                                                                                | .011                                                                                 | .779                                                                                  |
| Blind (RC+A)   |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |
| Seed           |  |  |  |  |  |  |  |  |
| 0              | .070                                                                                | .218                                                                                | .042                                                                                | .437                                                                                | .005                                                                                | 1.074                                                                               | .016                                                                                 | .558                                                                                  |
| 1              | .054                                                                                | .300                                                                                | .044                                                                                | .383                                                                                | .016                                                                                | .871                                                                                | .006                                                                                 | 1.032                                                                                 |
| 2              | .076                                                                                | .236                                                                                | .048                                                                                | .238                                                                                | .031                                                                                | .383                                                                                | .010                                                                                 | .979                                                                                  |
| 3              | .059                                                                                | .203                                                                                | .044                                                                                | .456                                                                                | .033                                                                                | .446                                                                                | .562                                                                                 | .841                                                                                  |
| Avg            | .065                                                                                | .239                                                                                | .045                                                                                | .378                                                                                | .021                                                                                | .693                                                                                | .148                                                                                 | .853                                                                                  |

## B DEMOGRAPHY

We compare the rates of extinction and of mining emergence as they depend upon world size and upon agents' access to global direction by way of the compass sensor. Extinction is defined as the population reaching zero individuals, whereas mining behavior is defined precisely as in the earlier experiments (a 10% decline in free biomass on land).

Runs are stratified according to both world size and compass access. We simulated 64 runs apiece of  $128 \times 128$  without compass and  $128 \times 128$  with compass, together with 32 runs apiece of  $512 \times 512$  without compass and  $512 \times 512$  with compass. For each of the four groups, we fit a non-parametric Kaplan-Meier estimator to the time-to-extinction distribution, treating those runs that did not go extinct within 2M steps as right-censored. In Figures 7 and 8 we observe that worlds tend both to survive longer and to produce mining more expeditiously in proportion as they are large or as their agents are furnished with compass information.

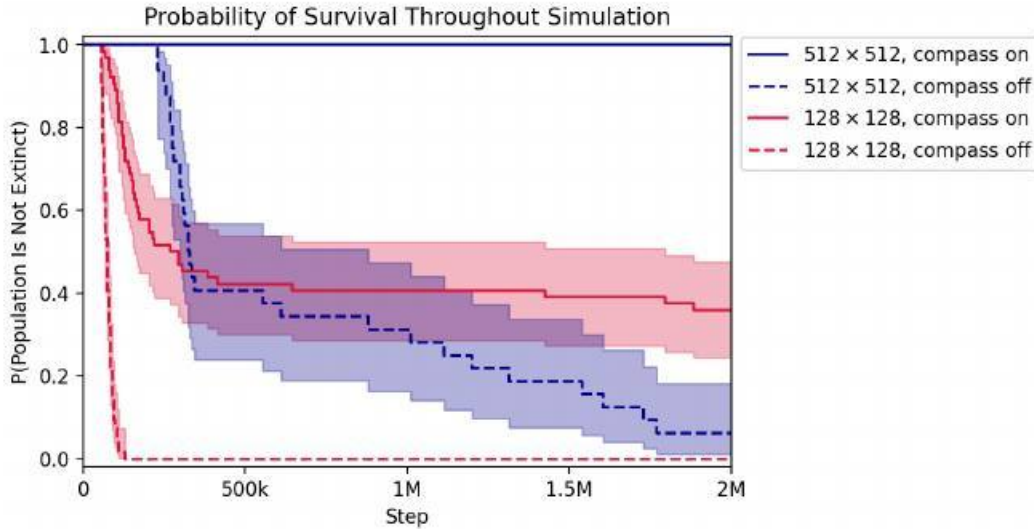


Figure 7: Probability that the population has not gone extinct, conditioned upon a given world size, compass access, and timestep. We fit a distinct non-parametric Kaplan-Meier estimator for each group of runs (e.g.  $512 \times 512$ , compass on). The shaded areas denote the 95% confidence intervals attending each curve. Each  $512 \times 512$  group fits the estimator upon 32 runs, while each  $128 \times 128$  group fits upon 64 runs.

We compare the timing of emergence and of extinction by means of a Cox proportional hazards model—a semi-parametric survival model that relates covariates to event hazard rates (Table 7). Runs that did not experience an emergence or extinction event by the conclusion of 2M time steps were treated as censored. In the case of extinction, both larger world size and compass use substantially attenuate the hazard; indeed, the extinction rate in  $128 \times 128$  worlds exceeds the corresponding rate in  $512 \times 512$  worlds by roughly a factor of 25 ( $HR=0.04$ ), while enabling the compass produces an effect of comparable magnitude ( $HR=0.03$ ). In the case of mining emergence, larger worlds engender mining behavior far more rapidly ( $HR=6.07$ ), and compass use accelerates emergence dramatically ( $HR=34.59$ ). The models exhibit sound discriminative capacity between groups (concordance 0.80 for extinction, 0.85 for mining). We conclude that larger worlds are able to sustain larger populations, thereby diminishing the likelihood of encountering the absorbing zero boundary, while concurrently exploring a more expansive phenotype space by virtue of the greater aggregate quantity of mutation relative to smaller worlds.

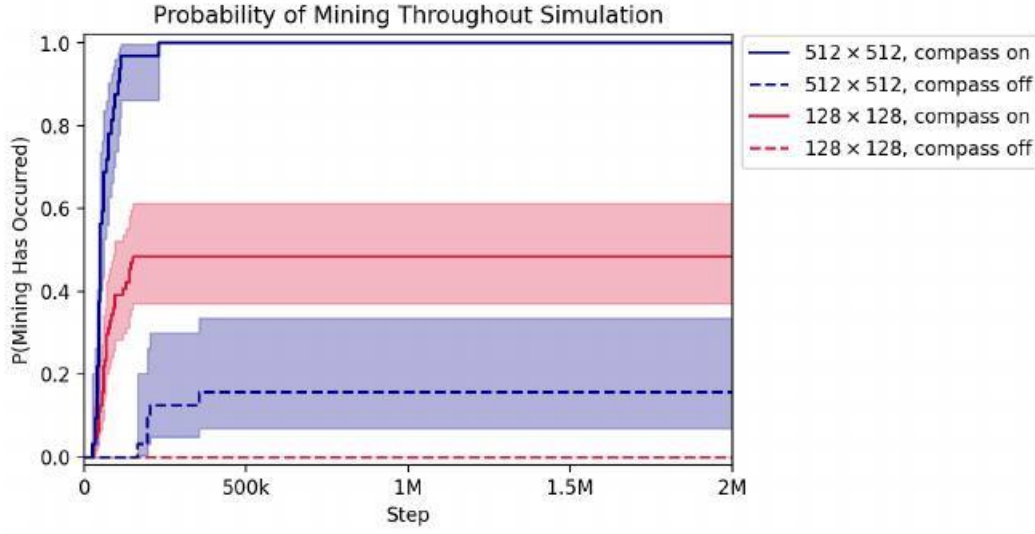


Figure 8: Probability that mining has evolved within the population, conditioned upon a given world size, compass access, and timestep. We fit a distinct non-parametric Kaplan-Meier estimator for each group of runs (e.g.  $512 \times 512$ , compass on). The shaded areas denote the 95% confidence intervals attending each curve. Each  $512 \times 512$  group fits the estimator upon 32 runs, while each  $128 \times 128$  group fits upon 64 runs.

Table 7: Cox proportional hazards models for extinction and emergence events, taking a  $128 \times 128$ , compass-off configuration as the baseline. Hazard ratios (HR) less than unity betoken a reduced hazard (and hence a longer time to event); by way of illustration, an extinction  $HR = 0.04$  for  $512 \times 512$  signifies that extinction transpires at 0.04 times the rate observed in  $128 \times 128$ . The 192 runs in total comprise 32 seeds with compass on and 32 seeds with compass off within the  $512 \times 512$  world, together with 64 seeds with compass on and 64 seeds with compass off within the  $128 \times 128$  world.

| Covariate             | Extinction |           |           | Mining   |             |           |
|-----------------------|------------|-----------|-----------|----------|-------------|-----------|
|                       | HR         | 95% CI    | p-value   | HR       | 95% CI      | p-value   |
| Size $512 \times 512$ | 0.04       | 0.02–0.07 | $< 0.005$ | 6.07     | 3.61–10.20  | $< 0.005$ |
| Compass On            | 0.03       | 0.01–0.05 | $< 0.005$ | 34.59    | 13.51–88.57 | $< 0.005$ |
| <b>Concordance</b>    | 0.80       |           |           | 0.85     |             |           |
| <b>Events / Total</b> | 135 / 192  |           |           | 68 / 192 |             |           |

## C ADDITIONAL FORAGING AND PREDATION RESULTS

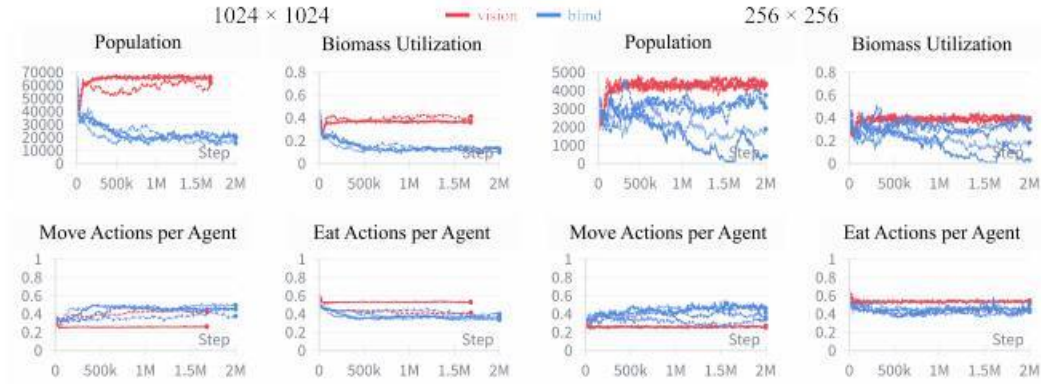


Figure 9: Foraging behaviors within the **Ocean** world at differing scales. Enabling vision in agents (red) issues in more stable populations and greater biomass utilization, a consequence of more frequent and more effective eating together with diminished movement. At larger scales, these dynamics prove more stable, exhibiting less variance across runs. It bears noting that, on account of limitations upon GPU hours, the vision seeds at the  $1024 \times 1024$  scale were run for 1.7M rather than 2M steps.

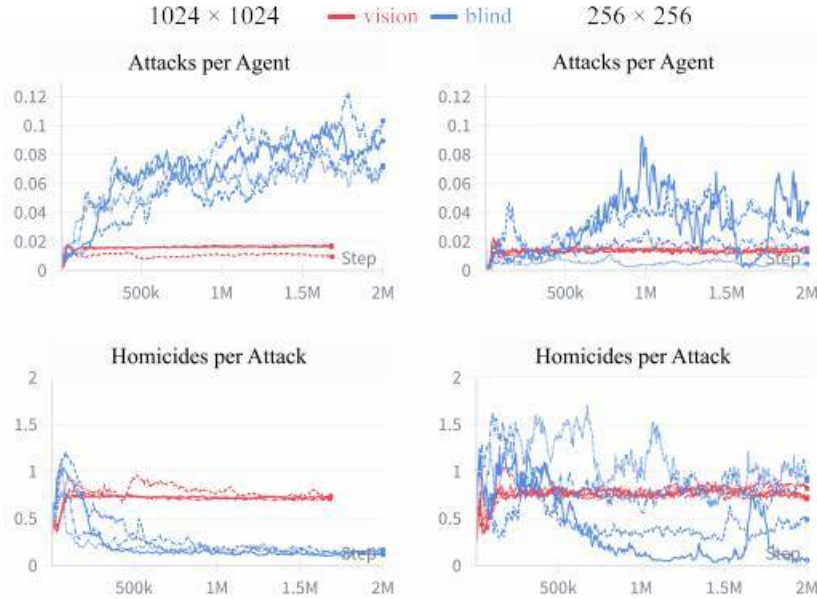


Figure 10: Predation behaviors within the **Ocean** world at differing scales. When endowed with vision (red), agents attack less frequently yet with considerably greater precision, a pattern suggestive of the emergence of specialized predation behaviors alongside foraging capabilities (Figure 9). This dynamic is once again more pronounced at larger scales, with greater variance manifesting in the behaviors of blind agents (blue) at smaller world sizes.

## D ADDITIONAL EXPERIMENT VISUALIZATIONS

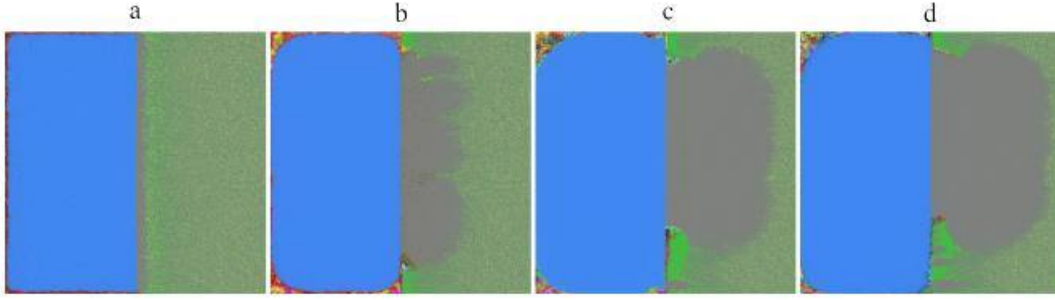


Figure 11: Larger-scale environments induce more varied and complex ecologies, which in turn give rise to distinctive visual patterns. Within this  $1024 \times 1024$  **Beach** world, (**RC**) agents initially perished at the shore (a), thereby transferring biomass to dry land. They subsequently developed mining behavior, clearing biomass from the beach (b). Once biomass had re-accumulated at the extremities of the beach (c), subpopulations of agents re-developed mining behavior, travelling inland along these miniature biomass beaches so as to extract the resources (d).

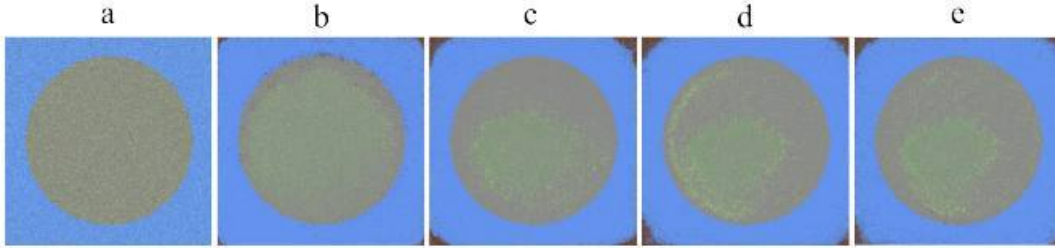


Figure 12: Mining behavior emerges across varied terrains. Within this  $512 \times 512$  **Island** world populated by (**RC**) agents, distinct regions of the shore are mined over the course of the run (b) (c). Biomass is transferred back to portions of the island once mining grows difficult and agents perish before they can return to water (d). Agents thereupon adapt so as to re-mine this wall of accumulated biomass, clearing it from the shore (e).

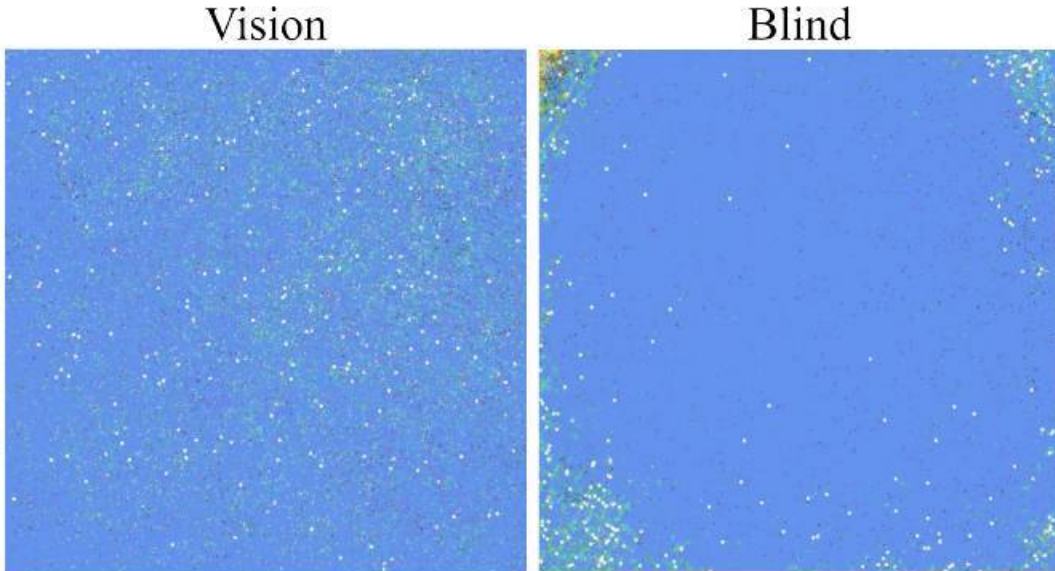


Figure 13: Representative runs within a  $512 \times 512$  **Ocean** world in which agents have attacking enabled: vision (**RCV+A**) agents at the left and blind (**RC+A**) agents at the right. The white squares signify that an agent is attacking at that location. When endowed with vision, agents adapt so as to locate free resources throughout the ocean world and to prey upon other agents more effectively (see Figures 5 and 6). Blind agents, by contrast, explore the ocean for resources to a far lesser extent, adopting instead policies that cluster them within the corners of the world.

## E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We set forth details pertaining to simulation design, runtime, and the reproducibility of our experiments.

**Simulator design.** We conduct the entirety of our experiments by means of a JAX-based, GPU-accelerated simulator devised for large-scale multi-agent ecosystems sustaining open-ended populations. Our simulator computes every constituent of the environment step upon the GPU—terrain dynamics, resource flows, agent updates, and measurement alike—so that policy evaluation and simulation share a single JAX compilation pipeline. The simulator exposes a minimal population-level interface: given the current map state, the agent states, and the heritable traits, a population algorithm supplies actions and updated traits and thereupon adjudicates births, deaths, and environmental transitions. This separation permits the same environment implementation to be reused in conjunction with differing evolutionary algorithms, reinforcement learning methods, or other population update rules. All operations are implemented as pure JAX functions combined with vectorization and just-in-time compilation, whereby they may be fused into a small number of GPU kernels. This affords the throughput requisite for our scaling experiments, with runtime measurements for a single H100 GPU reported in Table 8.

A salient feature of the simulator resides in its built-in tooling, which encompasses a measurement API for computing and recording aggregate statistics—biomass utilization, movement and eating rates, attack and homicide rates, and an extensive array of environment-level parameters at each step—that we employ directly in our figures and tables. In addition, an interactive 3D viewer renders terrain, water, resources, and agents from the generated reports and supports the inspection of individual agents (Figure 1). These tools proved indispensable for the discovery and validation of qualitative phenomena—such as long-range resource mining and predation patterns—that are not manifest from scalar metrics alone.

Viewed from a high-performance computing standpoint, our simulator fills a lacuna relative to existing simulators. General-purpose agent-based frameworks such as MASON (Luke et al., 2005), FLAME-GPU (Richmond et al., 2024), and Griddly (Bamford et al., 2020) afford considerable configurability, yet they demand substantial environment engineering and do not directly target the open-ended, evolving populations at the scales we investigate. Conversely, GPU-native multi-agent RL environments place their emphasis upon fixed numbers of agents and task-specific reward structures. We, by contrast, conjoin GPU-native execution, explicit support for births and deaths accompanied by mutable traits, and configurable resource and terrain dynamics, thereby enabling open-ended populations exceeding  $10^5$  agents while remaining interoperable with diverse population algorithms. This confluence yields a state-of-the-art platform for the scaling of ecological experiments such as those presented in this paper, as well as a serviceable foundation for future work on emergent behavior within large populations. Our simulator will be released upon acceptance.

**Runtime data.** Table 8 presents runtime data across differing grid sizes and maximum agent populations. Each run was conducted upon a single H100 GPU for a maximum of 2M steps. It bears noting that, although this table reports runtimes for (RC) and (RCV + A) agents possessed of 2 MLP layers and 64 channels per layer ( $\sim 22k$  parameters in total), the runtime varies with the size of the agent networks, the available sensors, and the level of logging.

Our largest runs at the  $1024 \times 1024$  grid size require  $\sim 10$  hours in the **Beach** terrain and  $\sim 14$  hours in the **Ocean** terrain on a single H100, with memory usage of roughly 60GB. **Ocean** runs prove marginally slower on account of the enabled agent attack action and the vision sensing.

**Reproducibility.** By reason of the large and randomized character of our environments, the precise outcomes of each experiment may differ substantially from one run to the next. Accordingly, we report the results obtained under four distinct random seeds for each experiment. Regrettably, owing to nondeterminism in certain low-level GPU operations, we find that repeating an experiment under an identical random seed does not reliably reproduce the same results. In particular, modern GPUs may execute parallel floating-point operations such as summations in differing orders across runs, and may fuse operations in ways that alter rounding behavior. Very small discrepancies arising from finite-precision arithmetic can, over the course of many steps, accumulate into a noticeable drift in the simulation trajectory. The use of 16-bit weights can coarsen these rounding effects, though it

Table 8: Runtime data within the **Beach** and **Ocean** worlds across differing grid sizes and maximum agent populations. Each **(RC)** agent is governed by a 22,465-parameter neural network, and each **(RCV + A)** agent by a 22,529-parameter neural network. We report simulation steps per second (hz) together with the total extrapolated wall-clock runtime in hours for 2M steps. The effective runtime is lower for those seeds in which populations go extinct prior to 2M steps.

|                 | 1024 × 1024 |          | 512 × 512 |         | 256 × 256 |         | 128 × 128 |         | 64 × 64 |         |
|-----------------|-------------|----------|-----------|---------|-----------|---------|-----------|---------|---------|---------|
|                 | 131072      |          | 32768     |         | 16384     |         | 8192      |         | 4096    |         |
| Environment     | hz          | runtime  | hz        | runtime | hz        | runtime | hz        | runtime | hz      | runtime |
| Beach (RC)      | 55          | 10.1 hrs | 190       | 2.9 hrs | 315       | 1.8 hrs | 535       | 1.0 hrs | 750     | 0.7 hrs |
| Ocean (RCV + A) | 40          | 13.9 hrs | 142       | 3.9 hrs | 235       | 2.4 hrs | 415       | 1.3 hrs | 750     | 0.7 hrs |

is not their primary source. Notwithstanding these run-to-run differences, clear high-level patterns emerge once the results of multiple seeds are taken in aggregate.

## F ABLATIONS

We present a series of ablations that interrogate the effects of specific hyperparameters and the robustness of our results under differing environment and agent configurations.

### F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We evaluate the effect that the initial population size within a 256×256 **Beach** world exerts upon the consistency of mining behavior and the stability of the population. The results reported pertain to **(RC)** agent populations. Agents bereft of a compass **(R)** exhibited a 0% mining rate and a 100% extinction rate across all initial population sizes save 8192, the highest tested, which yielded a 25% mining and 75% extinction frequency. It bears noting that the mining and extinction frequencies need not sum to 100%: it may transpire, for instance, that a population develops mining yet subsequently goes extinct, or that a population endures for the full 2M steps of our experiments without ever developing the mining behavior.

In Figure 14 we find that the initial population size exerts no appreciable effect upon mining consistency or extinction rates: populations commencing with a mere 128 agents evolve mining behaviors as frequently as those commencing with 8192 agents, and extinction rates remain largely invariant across the initial population size.

Figure 15 affords some insight into why so slight an effect is discernible. Irrespective of the magnitude of the initial population, each run converges rapidly to a comparable population size within the first 50,000 steps of simulation. This intimates that the environment implicitly stipulates a carrying capacity predicated upon the fixed resources available within the system: populations commencing below the carrying capacity grow so as to attain it, while those that are excessively large are constrained down to the capacity. In consequence, the effective population size proves comparable following an early correction of the initial population size, leaving the mining and extinction frequencies unperturbed. Importantly, however, larger initial population sizes contribute somewhat to the total resources within the system, inasmuch as agents carry biomass that is deposited upon death; accordingly, larger initial population sizes attain, on average, higher maximum populations over the course of simulation (Table 9), though the effect is not of sufficient magnitude to bear upon the mining or extinction frequency in this setting. It is worth noting that, although the population size converges early on, it does not thereafter remain constant throughout the run. Later in the simulation, the populations across distinct runs may diverge substantially, as the ecological consequences of agents evolving different abilities at different junctures reconfigure the subsequent resource distributions.

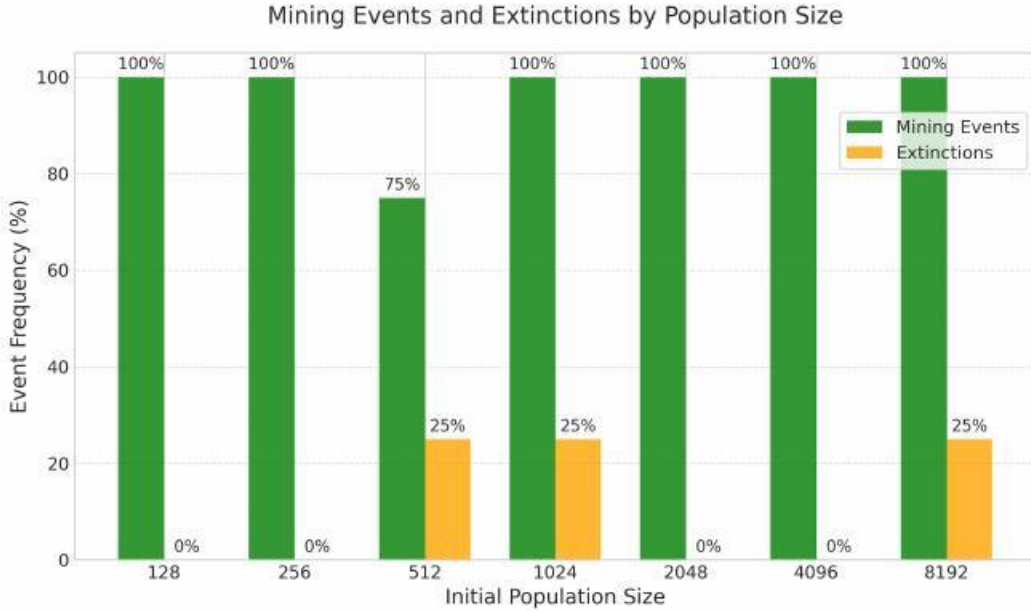


Figure 14: Frequency of mining and extinction events within a  $256 \times 256$  **Beach** world for differing initial sizes of (**RC**) agent populations. Frequencies are computed across four random seeds for each initial population size; that is, a 25% Extinction Event Frequency signifies that in 1 of 4 seeds the population declined to 0 prior to 2M simulation steps.

Table 9: The maximum population size attained after step 0 for differing starting population sizes, together with averages and standard deviations across four random seeds. Larger initial population sizes attain marginally higher maximum populations, with greater variance across seeds, than do lower initial population sizes. The maximum population attained is, however, largely robust to the initial population size: an initial population size (128) amounting to a mere 1.6% of an initial size of 8192 nonetheless attains 78.9% of the maximum population reached by the latter (3178 versus 4026).

| Initial Population Size | Max Population Size After Step 0 |
|-------------------------|----------------------------------|
| 128                     | $3178 \pm 217$                   |
| 256                     | $3056 \pm 61$                    |
| 512                     | $3451 \pm 437$                   |
| 1024                    | $3613 \pm 381$                   |
| 2048                    | $3786 \pm 431$                   |
| 4096                    | $3689 \pm 437$                   |
| 8192                    | $4026 \pm 642$                   |

## F.2 VARYING RESOURCE RULES

We evaluate the effect that two resource hyperparameters—the biomass photosynthesis rate and the initial biomass site density—exert upon the mining and extinction frequency. The setting is equivalent to that of Appendix F.1. For this study the initial population size is fixed at 2048, and agents are once again furnished with a compass (**RC**).

The biomass photosynthesis rate governs the rate at which landscape biomass creates energy via photosynthesis, thereby furnishing resources to agents over time, whereas the initial biomass site density parameter determines the fraction of grid cells endowed with biomass at the outset of simulation. All experiments in the main body (Section 4) adopt a biomass photosynthesis rate of 0.01 and an initial biomass site density of  $1/4$ .

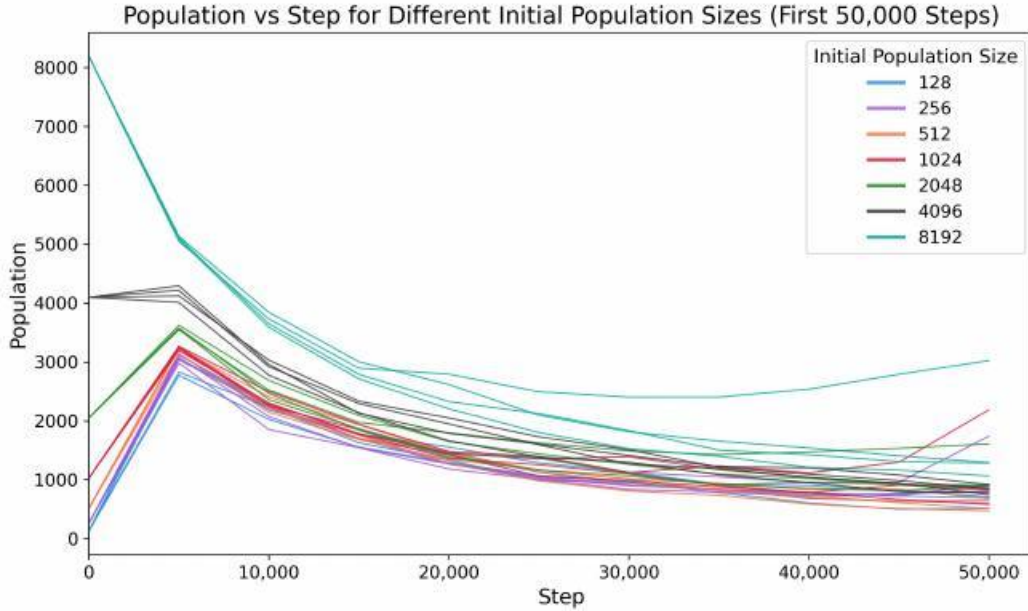


Figure 15: Population size over the first 50,000 time steps within the  $256 \times 256$  **Beach** populated by (RC) agents, with four seeds per initial population size. The environment sustains an implicit carrying capacity predicated upon the resources available within the system, and all initial population sizes approach this value early in the run. Larger initial population sizes contribute to the total resources within the system by virtue of the biomass deposited upon agent death, thereby raising the carrying capacity somewhat.

Whereas varying the initial population size did not affect mining and extinction consistency (Figure 14), altering the resource rules exerts a pronounced effect, as demonstrated in Figures 16 and 17

Intuitively, larger biomass photosynthesis rates or initial biomass site densities directly augment the effective carrying capacity of the environment by increasing the resources available within the system. In consequence, larger populations are sustained throughout the run, which heightens the likelihood of mutations conducive to mining behavior while diminishing the prospect of extinction. Our data corroborate this intuition: on average (omitting those seeds that went extinct), runs with an initial biomass site density of  $1/16$  attained a maximum population of 1608 after step 0, in contrast to a mean maximum population of 2466 for density  $1/8$ , 3786 for density  $1/4$ , and 5353 for density  $1/2$ .

### F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

One might raise the concern that the heightened stability of mining behavior within large environments (Table 1) admits of explanation purely by self-averaging rather than by qualitatively distinct dynamics. By self-averaging, we invoke the statistical-physics intuition that, as system size grows, the fluctuations in macroscopic quantities contract roughly as  $1/\sqrt{N}$ —where  $N$  denotes the system size—such that a single large system comports itself like the average of smaller systems of the same aggregate size. On this construal, a  $512 \times 512$  **Beach** world might amount to nothing more than many loosely coupled  $128 \times 128$  worlds run in parallel. Were this the case, then our scaling results could in principle be reproduced simply by running a sufficient number of small worlds and averaging over random seeds.

To adjudicate this hypothesis, we conduct an experiment that holds the total “sample size” approximately fixed while varying the manner in which it is partitioned across environments. Within the **Beach** terrain, populated by compass-equipped agents (RC), we consider three conditions:

- 4 runs of a  $512 \times 512$  world with 8192 initial agents each,

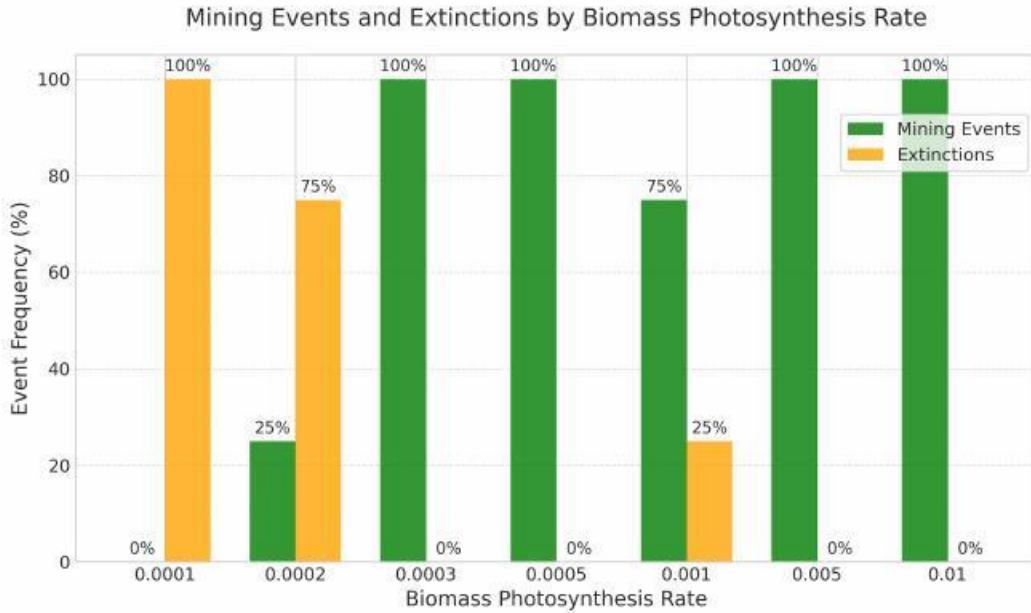


Figure 16: Frequency of mining and extinction events within a  $256 \times 256$  **Beach** world for differing settings of the biomass photosynthesis rate resource parameter. Frequencies are computed across four random seeds for each setting. Mining emerges consistently at biomass photosynthesis rates of 0.0003 and above.

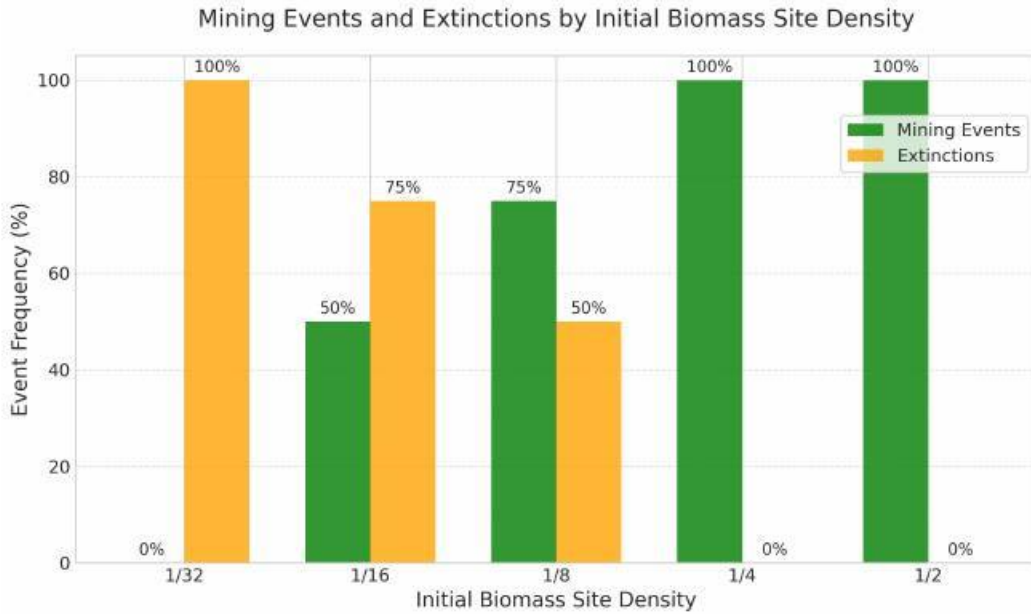


Figure 17: Frequency of mining and extinction events within a  $256 \times 256$  **Beach** world for differing settings of the initial biomass site density resource parameter. Frequencies are computed across four random seeds for each setting. Greater initial biomass site densities issue in more consistent discovery of mining and a diminished prospect of extinction.

- 16 runs of a  $256 \times 256$  world with 2048 initial agents each,
- 64 runs of a  $128 \times 128$  world with 512 initial agents each.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps under the same hyperparameters as in Section 4.1. By way of control, we replicate this protocol for agents devoid of a compass (**R**). Table 10 summarizes the frequency of long-distance mining and of extinction within each configuration.

Were large environments merely functioning as self-averages of many smaller ones, we should expect mining and extinction rates to remain roughly invariant to the manner in which we apportion grid cells and agents across runs: 4 large worlds or 64 small ones would evince comparable probabilities of discovering mining. Instead, we observe a pronounced dependence upon world size. For (**RC**) agents, mining emerges in all runs, with no extinctions whatsoever, at  $512 \times 512$ ; it remains common though not guaranteed at  $256 \times 256$ ; and it grows unreliable and fragile at  $128 \times 128$ , wherein many populations go extinct before attaining 2M steps. (**R**) agents, by contrast, never develop mining and succumb to extinction at every scale, thereby confirming that the compass remains a pivotal enabler of long-range behavior.

Table 10: Mining and extinction events under a trade-off of world size against the number of independent runs. We compare **Beach** worlds at grid sizes  $512 \times 512$ ,  $256 \times 256$ , and  $128 \times 128$ , with initial populations scaled proportionally (8192, 2048, and 512 agents, respectively). For each size, we apportion the same total number of grid cells and initial agents across all runs by running 4 seeds at  $512 \times 512$ , 16 seeds at  $256 \times 256$ , and 64 seeds at  $128 \times 128$ . We report the fraction of runs in which mining emerged (🔍) together with the fraction in which the population went extinct before 2M steps (💀), for agents endowed with a compass (**RC**) and for those without one (**R**).

|               | $512 \times 512$ |     | $256 \times 256$ |       | $128 \times 128$ |       |
|---------------|------------------|-----|------------------|-------|------------------|-------|
|               | 8192             |     | 2048             |       | 512              |       |
|               | 🔍                | 💀   | 🔍                | 💀     | 🔍                | 💀     |
| ( <b>RC</b> ) | 4/4              | 0/4 | 15/16            | 1/16  | 31/64            | 41/64 |
| ( <b>R</b> )  | 0/4              | 4/4 | 0/16             | 16/16 | 0/64             | 64/64 |

These results are irreconcilable with a self-averaging explanation. On the contrary, larger environments appear to enable and to stabilize mining behavior in ways that cannot be accounted for solely by averaging over a greater number of small simulations.

Even when the total number of cells, agents, and simulation steps is matched across conditions, the concentration of resources into a few large worlds produces a regime in which mining is consistent and extinctions rare, whereas the dispersal of the same resources across many small worlds yields populations that frequently fail to discover mining and are more prone to collapse. This intimates that increasing the environmental scale does not merely smooth out stochastic noise, but rather alters the underlying ecological dynamics—for instance, by sustaining larger and more spatially distributed populations together with a more persistent diversity among evolving policies.

#### F.4 MUTATION RATE SWEEP

We probe the sensitivity of the emergence of mining to the mutation rate hyperparameter, which governs the scale of the Gaussian noise added to a child agent’s policy upon birth. Intuitively, the mutation rate directly conditions the degree to which a child agent diverges from its parent. We once again conduct experiments within a  $256 \times 256$  **Beach** world, with an initial population of 2048 (**RC**) agents, for a maximum of 2M steps.

In Figure 18 we show that, between mutation rates of  $6 \times 10^{-3}$  and  $7.875 \times 10^{-2}$ , mining emerges in at least 3 of the 4 seeds run, which indicates that our results in Section 4.1 at mutation rate  $3 \times 10^{-2}$  are robust throughout this interval. As anticipated, mutation rates that are excessively small (e.g.,  $3 \times 10^{-3}$ ) preclude agents from developing new behavior and impair their survival, thereby precipitating extinction. Conversely, large mutation rates (e.g.,  $3 \times 10^{-1}$ ) subvert population stability, consistently precipitating extinction in the absence of any mining adaptation.

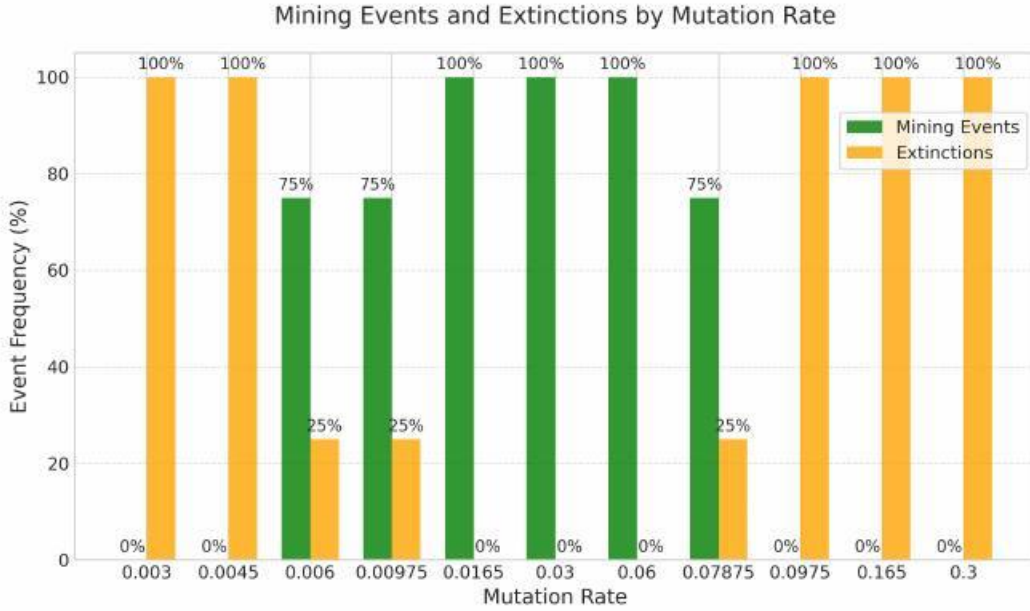


Figure 18: Frequency of mining and extinction events within a  $256 \times 256$  **Beach** world for differing mutation rates. Frequencies are computed across four random seeds for each setting.

## F.5 NETWORK SIZE

We study the effect that the size and architecture of the agent policy network exert upon the mining and extinction frequency, as well as the manner in which environmental scale modulates this effect. We investigate  $256 \times 256$  and  $512 \times 512$  **Beach** worlds with initial population sizes of 2048 and 8192 (**RC**) agents, respectively (scaling by a factor of four with the number of grid cells). We ran baselines with (**R**) agents for each experiment, once again encountering rare-to-nonexistent mining events and 100% extinction rates.

The main experiments in Section 4 employed agent policies possessed of 2 layers and 64 channels, yielding a network of 22465 parameters per agent (Table 11). In Figure 19 we show that, within the  $256 \times 256$  **Beach**, halving the channels to 32 (which yields 9185 parameters in total) or doubling them to 128 (which yields 61313 parameters), with the number of layers held fixed, exerts minimal-to-no effect upon the mining and extinction frequency. Altering the network depth, by contrast, exerts a pronounced effect: all populations of agents endowed with a mere 1 layer and 32 channels went extinct, as did 50% of the populations of agents endowed with 3 layers and 128 channels.

Crucially, scaling the environment up to the  $512 \times 512$  grid size produced markedly different dynamics for networks of varying depth (Figure 20). In particular, within the larger world, agent populations endowed with 3-layer, 128-channel networks mined consistently and survived through 2M steps. Surprisingly, agent populations endowed with the small 1-layer, 32-channel networks mined consistently as well. We conjecture that the discovery of new behaviors poses a greater challenge to both smaller and larger networks, albeit for divergent reasons. Where the network is smaller, its reduced capacity renders good solutions difficult to locate; where the network is larger, the enlarged search space entails that promising new behavior takes longer to discover. Whatever the underlying reasons, the fact that these networks fail so frequently within small environments yet succeed within larger ones further illustrates the stability that accrues from increasing scale. The larger environment can sustain larger populations, which affords more extensive exploration of policy space; concomitantly, locally successful mutants take longer to propagate across a large environment, which fosters diversity and, once again, ensures more extensive exploration. For further discussion of these effects, see (Mitchell et al. 2006).

Table 11: Total parameters in **(RC)** and **(R)** agent policies across differing network architectures.

| Architecture<br>(layers, channels) | Parameter Count |            |
|------------------------------------|-----------------|------------|
|                                    | <b>(RC)</b>     | <b>(R)</b> |
| 1, 32                              | 8129            | 7969       |
| 2, 32                              | 9185            | 9025       |
| 2, 64                              | 22465           | 22145      |
| 2, 128                             | 61313           | 60673      |
| 3, 128                             | 77825           | 77185      |

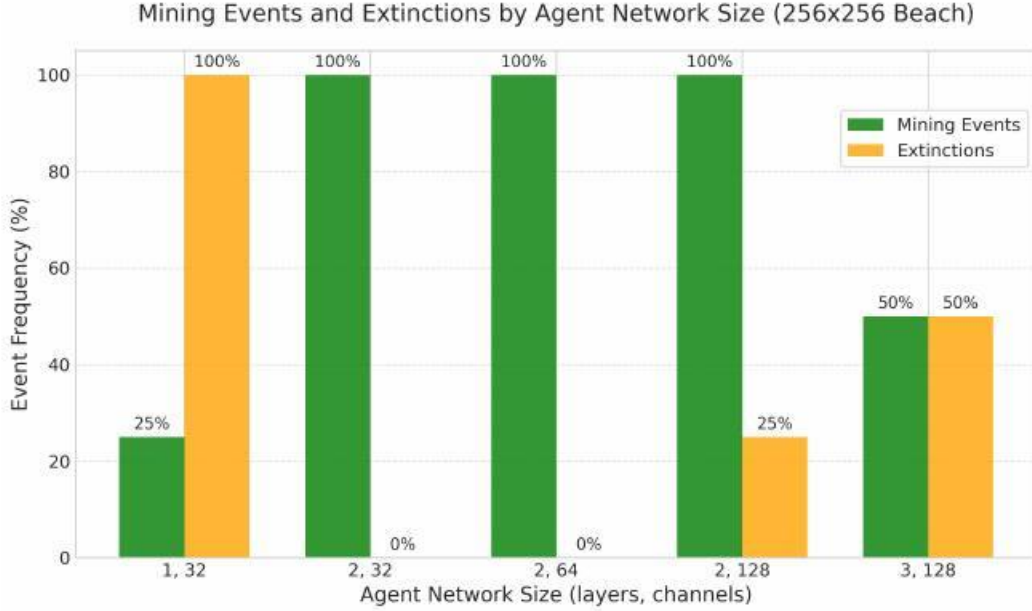


Figure 19: Frequency of mining and extinction events within a  $256 \times 256$  **Beach** world for differing agent policy network sizes. Frequencies are computed across four random seeds for each setting.

## F.6 VARYING VISION RANGE

We assess the impact that differing vision ranges and viewing configurations exert upon predation and foraging behaviors within the  $512 \times 512$  **Ocean** world populated by **(RCV+A)** agents. We evaluate visual sensor ranges of  $3 \times 3$ ,  $7 \times 7$ , and  $9 \times 9$ . For each range, we evaluate the two viewing configurations depicted in Figure 21: a) Around, in which agents can perceive grid cells on every side of them; and b) Front, in which agents cannot perceive cells behind them. The total number of cells within the agent’s vision range is equal across the two configurations. The main experiments involving vision in Section 4.2 were conducted under the  $7 \times 7$  Around setting.

These results indicate that a smaller  $3 \times 3$  vision range issues in behavior largely indistinguishable from the absence of vision altogether (Blind), whereas the results at  $9 \times 9$  appear comparable to those at  $7 \times 7$ . This intimates that larger vision ranges may be subject to diminishing returns under the present environmental configuration—an outcome that is intuitive in light of the remainder of that configuration; for, inasmuch as the agents possess no memory, it is improbable that visual features situated at the periphery of a  $9 \times 9$  window could meaningfully alter the optimal action relative to a  $7 \times 7$  window with sufficient consistency to furnish a strong learning signal.

The data further suggest that, at the  $7 \times 7$  and  $9 \times 9$  vision ranges, the Around configuration proves marginally more favorable with respect to population stability and the effectiveness of the attack and eat actions, thereby yielding higher average population  biomass utilization , and homicides

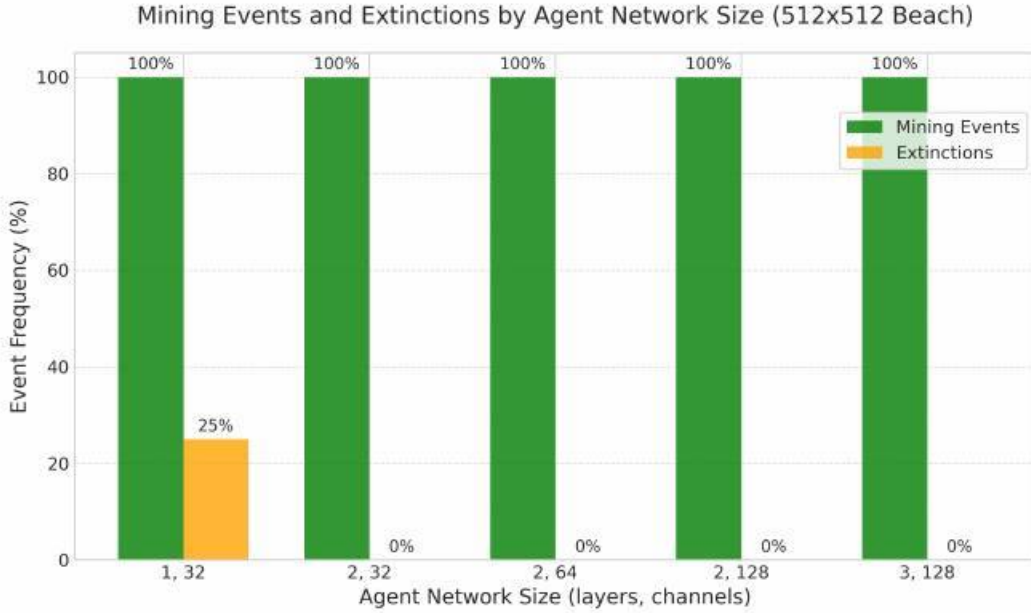


Figure 20: Frequency of mining and extinction events within a  $512 \times 512$  **Beach** world for differing agent policy network sizes. Frequencies are computed across four random seeds for each setting.

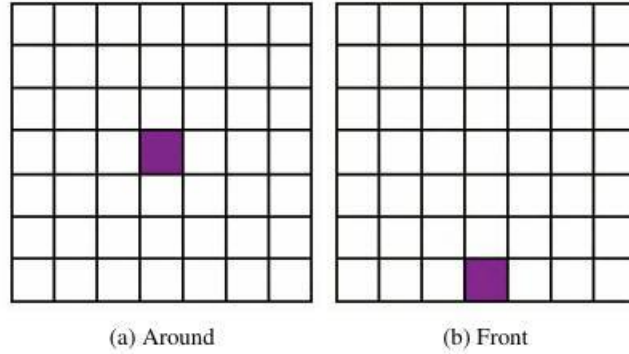


Figure 21: Comparison of the Around and Front viewing configurations for the  $7 \times 7$  vision range. The purple cell denotes an agent, oriented toward the top of the page.

per attack  than the Front configuration. We conjecture that this advantage owes to the diminishing returns attendant upon perceiving more distant pixels; for, in the Around configuration, a greater proportion of the pixels within the agent’s vision range lie nearby, and hence in all likelihood convey more information germane to near-term actions than do the distant cells visible under the Front configuration. At the  $3 \times 3$  vision range, however, the trend is inverted, the Front configuration proving more favorable than Around; this indicates that, within this environment, the marginal benefit of perceiving one additional pixel ahead through the  $3 \times 3$  Front configuration outweighs that of perceiving one pixel behind in the  $3 \times 3$  Around configuration, whose behavior most nearly approximates that of the Blind setting.

## F.7 VARYING ATTACK STRENGTH

We study the effect that reducing the agent attack strength exerts upon emergent behaviors and population dynamics within the  $256 \times 256$  **Ocean** world. We evaluate the following attack strengths: 1) Full Attack, which deals 10 HP of damage, instantly killing any agent within the attack radius; 2)

Table 12: Foraging and predation data within the  $512 \times 512$  **Ocean** world for vision agents (**RCV+A**) across varying vision ranges and viewing configurations. Refer to Tables 5 and 6 for the definitions of the metrics denoted by each icon. For each seed, the value of each metric was averaged across all steps commencing at step 500k, so as to permit them to stabilize. The reported values are the mean across the four seeds together with the standard deviation of the four seed averages.

| Vision Range | Viewing Configuration |  |  |  |  |  |  |
|--------------|-----------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 3×3          | Blind                 | 7212 ± 2496                                                                       | .181 ± .060                                                                       | .435 ± .090                                                                       | .392 ± .033                                                                         | .032 ± .006                                                                         | .307 ± .090                                                                         |
|              | Around<br>Front       | 10308 ± 4016                                                                      | .258 ± .105                                                                       | .352 ± .014                                                                       | .427 ± .022                                                                         | .040 ± .030                                                                         | .388 ± .195                                                                         |
|              |                       | 13863 ± 520                                                                       | .351 ± .023                                                                       | .336 ± .008                                                                       | .437 ± .009                                                                         | .026 ± .006                                                                         | .470 ± .145                                                                         |
|              | 7×7                   | 16739 ± 122                                                                       | .373 ± .003                                                                       | .259 ± .000                                                                       | .532 ± .001                                                                         | .015 ± .000                                                                         | .751 ± .007                                                                         |
|              |                       | 15255 ± 623                                                                       | .343 ± .015                                                                       | .253 ± .001                                                                       | .540 ± .004                                                                         | .015 ± .000                                                                         | .690 ± .041                                                                         |
|              | 9×9                   | 16744 ± 117                                                                       | .372 ± .002                                                                       | .257 ± .001                                                                       | .534 ± .001                                                                         | .015 ± .000                                                                         | .728 ± .009                                                                         |
|              |                       | 15211 ± 1416                                                                      | .342 ± .033                                                                       | .253 ± .001                                                                       | .541 ± .007                                                                         | .015 ± .000                                                                         | .664 ± .050                                                                         |

Half Attack, which deals 5 HP of damage; and 3) Quarter Attack, which deals 2.5 HP of damage. The main **Ocean** experiments in Section 4.2 were conducted under the Full Attack setting.

We find that reducing the attack strength from Full Attack to Half Attack substantially diminishes both the effectiveness and the frequency of attacks: attacks per agent and homicides per attack alike decline considerably (Figure 22). In consequence, populations subject to lower attack strength grow larger, their agents moving and eating more frequently and thereby attaining greater biomass utilization. Reducing the strength further to the Quarter Attack setting yields diminishing returns, both in the further curtailment of violence and in the facilitation of additional population growth. Intuitively, when attacks inflict less damage, a homicide demands a greater number of attacks per agent, which reduces the lethality of each individual attack. Yet once the lethality is thus reduced, attacking behavior becomes less useful and hence less liable to be selected for within the agent population. Agents subject to lower attack strengths accordingly do not become as adept at attacking, and instead devote more time to foraging for food.

## F.8 VARYING AGENT HP

We investigate the effect of reducing the starting HP of child agents and lowering the healing rate, so as to impose upon agents a pressure to ‘grow’ over the course of their lifetime in order to survive. The healing rate parameter determines the quantity of HP an agent can regenerate per healing step, a process that converts available energy into HP; agents can influence their healing only by maintaining sufficient energy.

We study two settings: 1) Full HP, in which child HP begins at 5 and the healing rate is 1 HP per 0.1 energy; and 2) Low HP, in which child HP begins at 1 and the healing rate is 0.5. The maximum agent HP is 10 across all settings. The main experiments (Section 4) were conducted under the Full HP setting.

Across the two settings there is no appreciable difference in homicides per attack, in eat action frequency, or in biomass utilization (Figure 23). Under the Full HP setting, populations grow marginally larger, whereas under the Low HP setting, agents attack less frequently and move somewhat more. We conjecture that agents born with lower HP must concentrate upon foraging early in their lifetime in order to attain full size, which would account for this difference.

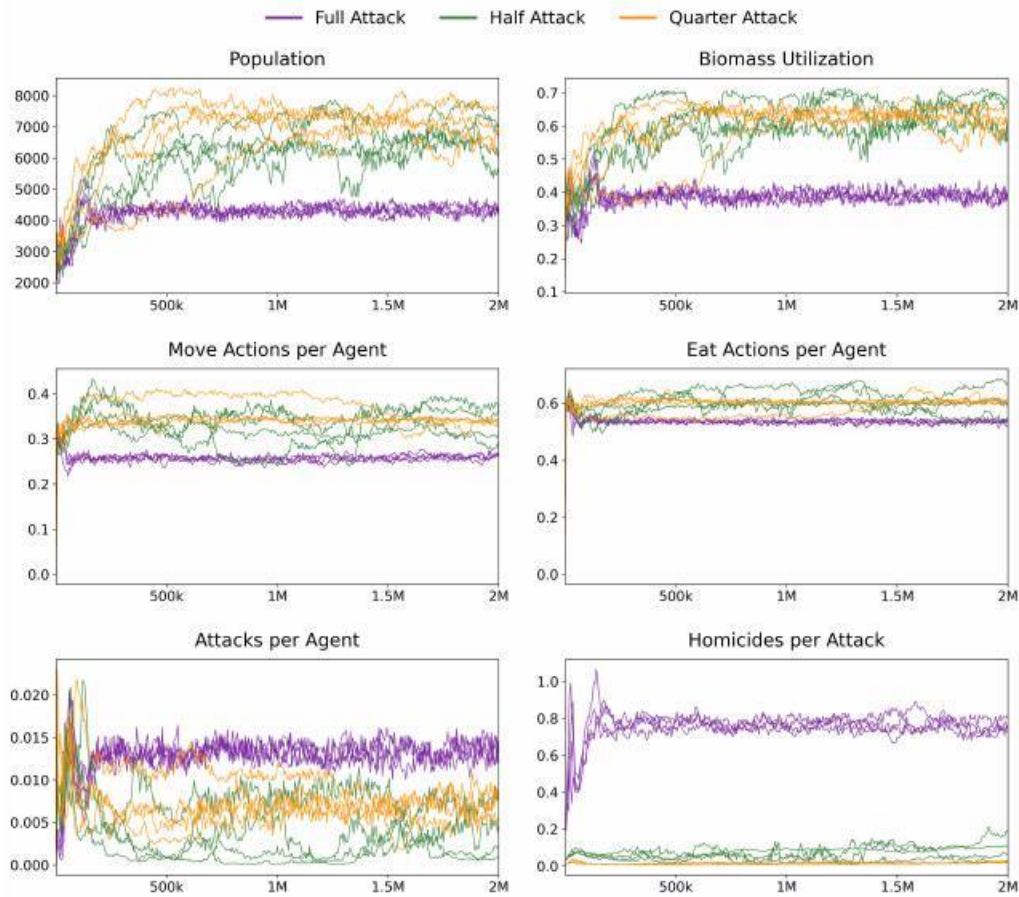


Figure 22: Population and action frequency data over 2M steps within a  $256 \times 256$  **Ocean** world for **(RCV+A)** agents across the Full Attack, Half Attack, and Quarter Attack strength settings. Four seeds are plotted for each attack strength setting.

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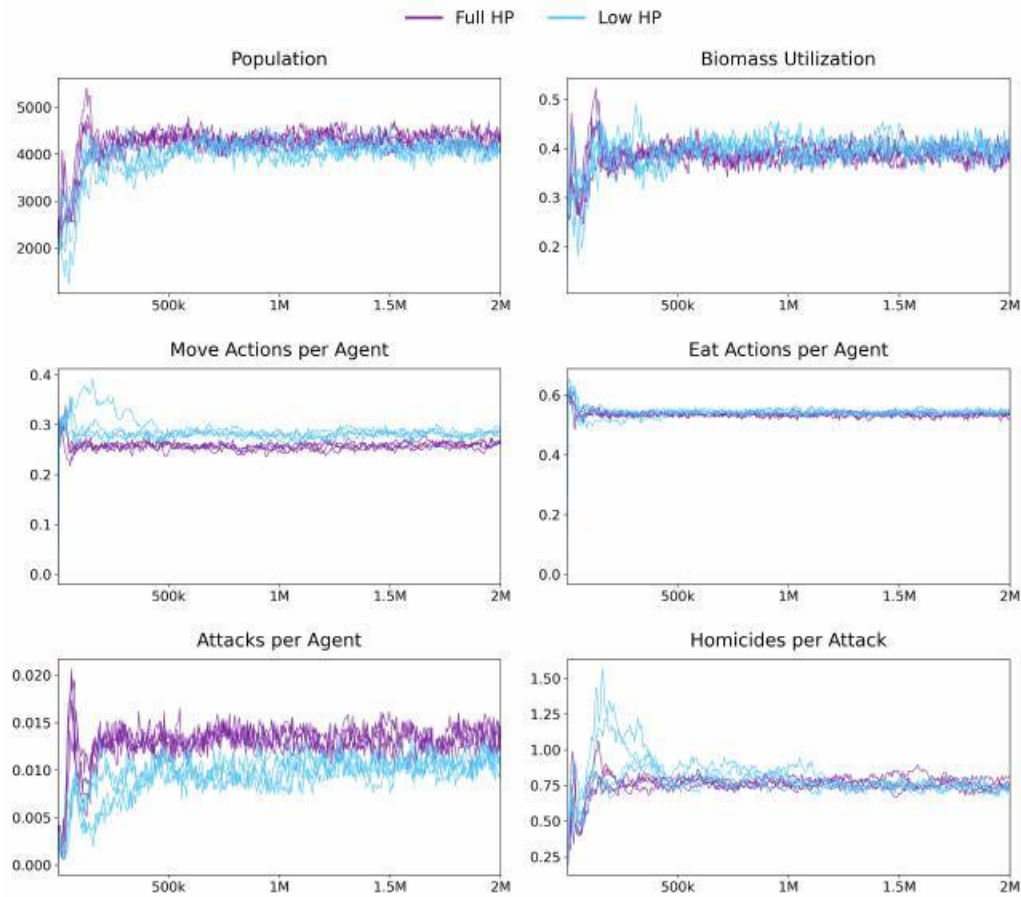


Figure 23: Population and action frequency data over 2M steps within the  $256 \times 256$  **Ocean** world for (RCV+A) agents across the Full HP and Low HP settings. Four seeds are plotted for each HP setting.